DRL-Based Single-Input Pinning Synchronization of Hopfield Chaotic Neural Networks with an Application to Image Encryption

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ABSTRACT

With a single bounded actuator, chaotic synchronization depends not only on where the input is applied but also on which variables are available to the controller. This study examines this relationship in a continuous-time Hopfield network using proximal policy optimization (PPO) as a common feedback-learning method. A complete comparison of the three paired four-dimensional subsets at two actuator locations shows that observation composition matters alongside actuator placement: replacing one four-dimensional subset markedly improves Node 2, but increases residuals and control effort at Node 3. An accuracy-constrained screening procedure translates these observations into candidate configuration choices and checks them on held-out initial conditions. Evaluation of all actuator subsets then quantifies the accuracy and control-effort consequences of restricting actuation. Favorable multi-input configurations improve full-horizon accuracy relative to single-input Node 3, but do not uniformly improve steady-state precision. Comparisons with fixed-gain feedback, linear quadratic regulation, sliding-mode control, and soft actor-critic reveal condition-dependent performance and effort tradeoffs rather than a universal PPO advantage. Local indicators support candidate screening; analytical state boundedness and post-training finite-time and multi-step sensitivity checks address distinct properties of the sampled closed loop. Synchronized chaotic trajectories are also used in an image encryption and decryption scheme combining Logistic-map masking and dynamic DNA coding, with statistical analyses indicating reduced image redundancy. The control results provide a network-specific assessment of actuator-dependent observation sensitivity and the performance cost of limited actuation.

1. Introduction

Chaotic synchronization concerns the coordination of sensitive nonlinear trajectories through coupling or feedback, with applications in secure communication and coordinated dynamical systems [1–4]. In a drive–response architecture, the control must make a response system follow an independently evolving drive. When only one actuator is available, a scalar input must influence all synchronization-error components through the internal coupling. The actuator location therefore becomes part of the control design: a controller that performs well at one coordinate may require different feedback information or control effort at another.

Linear feedback, adaptive control, sliding-mode control, and pinning strategies provide established approaches to synchronization [5–9]. These methods supply analytical structure and, under their respective assumptions, stability results. Deep reinforcement learning offers a complementary route by fitting nonlinear feedback from observations and rewards [10–12]. Applications to chaos control and synchronization have demonstrated the feasibility of this approach across different dynamical systems [13–18]. This motivates examining what learned feedback can achieve under a prescribed actuator constraint, while retaining model-based controllers as informative references.

Observation availability is a separate design variable. A single actuator does not require that the controller observe only the actuated coordinate: sensing determines the available information, whereas actuation determines where feedback enters the dynamics. Recent work has examined reinforcement learning with reduced inputs and incomplete observations [19, 20]. In particular, Weissenbacher et al. [20] study reduced sensing and memory architectures in chaotic-flow control. The present study instead examines how the location of a fixed scalar actuator affects the performance of a memoryless synchronization policy as its observation is reduced. This distinction motivates assessing actuator placement and observation availability together. Ozan et al. [21] also examine sensor and actuator locations in partially observed chaotic-flow control. Thus, considering sensing and actuation jointly is not itself claimed as new; the present focus is their measured consequence for single-coordinate drive–response synchronization in the specified Hopfield network.

Learned synchronization with incomplete state information has already been demonstrated for identical and distinct chaotic systems [15, 16]; partial observation alone

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is therefore not the contribution claimed here. Related soft-computing approaches include critic-based neuro-fuzzy chaos control [22], Lyapunov-analyzed adaptive fuzzy synchronization [23], and Lyapunov-based reward shaping for deep reinforcement learning in tracking control [24]. These methods motivate distinguishing learning performance from guarantees established for a particular controller structure.

The continuous-time Hopfield-type network provides a compact setting for this investigation. Unlike the symmetric network associated with equilibrium-seeking dynamics [25], asymmetric Hopfield-type connections can produce chaotic behavior [26–28]. Existing work includes dynamical analysis and synchronization using model-based, impulsive, intermittent, and event-triggered control [29–33]. The three-state model considered here permits examination of every actuator subset and direct comparison of local dynamical indicators with nonlinear closed-loop behavior. Its role is to make the constrained-control design choices inspectable; extension to larger networks requires separate evaluation.

The central question is how actuator location changes the consequences of restricting feedback information under bounded single-input synchronization. A second question concerns the accuracy and applied-effort cost of using fewer control channels. PPO supplies a common implementation for these comparisons, rather than being introduced as a new learning algorithm. Local transverse and linear-control indicators guide candidate screening; observed closed-loop behavior is then evaluated separately. State boundedness and post-training sampled-map sensitivity checks do not by themselves guarantee vanishing synchronization error.

The experiments connect each question to a corresponding comparison: node-dependent observation reduction examines feedback-information sensitivity, while all nonempty actuator subsets assess actuation-resource choices. Traditional controllers and matched-budget SAC determine how the findings depend on controller selection. The improvement over local-error fixed-gain feedback at Node 2 is one reference comparison, not evidence that conventional feedback cannot synchronize this channel. Error and applied effort are reported together.

Synchronized chaotic trajectories are additionally used in an image encryption and decryption demonstration based on Logistic-map masking and dynamic DNA coding. This application examines the use of the generated trajectories beyond the control task; its statistical image results are evaluated separately from the synchronization evidence.

The main contributions are as follows:

- A fixed-dimension assessment covers all three paired observation subsets at Nodes 2 and 3, separating input size from coordinate composition in the tested Hopfield network. Changing the four-dimensional observation subset improves synchronization at Node 2 but worsens residuals and effort at Node 3 under the same memoryless learning setup. An accuracy-constrained selection procedure then ranks feasible candidates by feedback dimension and control effort, with choices checked on held-out initial conditions. This provides a candidate-limited design assessment rather than a minimum sensing requirement or a new partial-observation algorithm.
- An evaluation of every nonempty actuator subset quantifies the full-horizon accuracy and total applied-effort costs of restricting actuation. Combined with traditional-controller and matched-budget SAC references, it distinguishes actuator availability from controller-specific performance and shows why additional inputs need not improve every measured criterion under a fixed training budget.

These empirical contributions are supported by an analytical state-boundedness result and numerical sensitivity checks for the deployed sample-and-hold policies. The analysis does not claim PPO training convergence, a certified synchronization-error bound, or a general optimal node-selection rule.

The remainder of this paper is organized as follows. Section 2 introduces the continuous-time Hopfield chaotic neural network and formulates the single-input pinning synchronization problem. Section 3 presents the local pinning-node selection analysis and the DRL-based control framework implemented by PPO. Section 4 reports the experimental setup, results, and discussion. Section 5 presents the application in image encryption. Finally, Section 6 concludes this paper and outlines possible future work.

2. Preliminaries

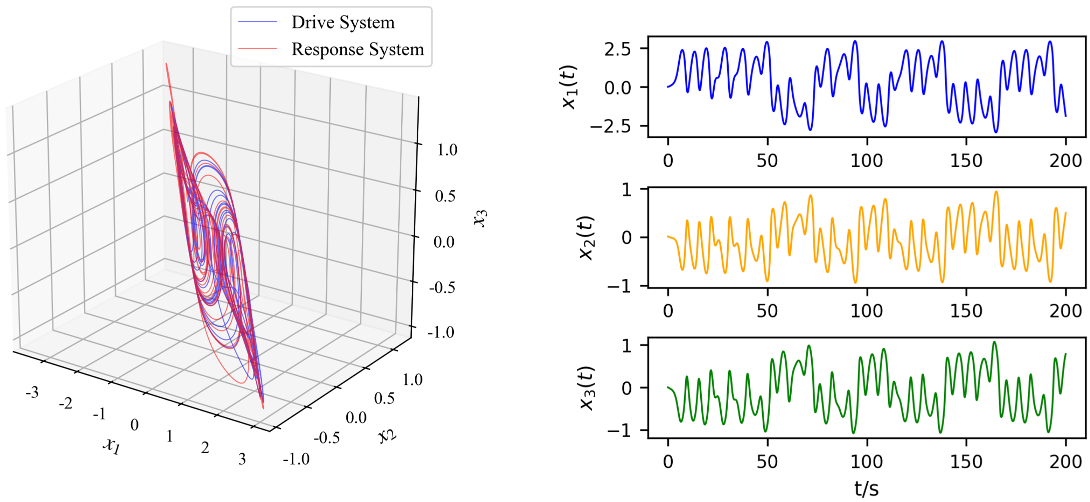
Hopfield's continuous neural network model proposed in 1984 [25] can be written as

$$C_i \frac{du_i}{dt} = \sum_{j=1}^{n} T_{ij} V_j - \frac{u_i}{R_i} + I_i, \tag{1}$$

with the static nonlinear input–output mapping $V_i = g_i(u_i)$. Here, u_i denotes the internal state variable of the ith neuron, V_i is the neuron output, C_i and R_i represent the equivalent capacitance and resistance, respectively, T_{ij} denotes the synaptic connection weight, and I_i is the external bias input.

In Hopfield's original study, the matrix T was assumed to be symmetric so as to guarantee the monotonic decrease of the system energy function in a mathematical sense. However, such a setting leads the network to eventually converge to a static equilibrium point and thus prevents the emergence of chaotic behavior. To focus on the nonlinear bifurcation and chaotic attractor characteristics induced by the complex internal topological coupling of the neural network, the original equations are often nondimensionalized in the control literature [26, 29].

Introducing the parameter substitutions $a_i = \frac{1}{R_i C_i}$, $w_{ij} = \frac{T_{ij}}{C_i}$, and $b_i = \frac{I_i}{C_i}$, and assuming that all neurons share the same leakage coefficient ($a_i = 1$), neglecting the constant bias term ($\mathbf{b} = \mathbf{0}$), and choosing $\tanh(\cdot)$ to characterize the saturating nonlinearity of the neuron output, one obtains the



(a) Phase-space trajectories

(b) Time-domain trajectories

Figure 1: Uncontrolled phase-space trajectories of the drive and response Hopfield chaotic neural networks and time-domain state trajectories of the continuous-time Hopfield-type chaotic neural network.

continuous-time Hopfield-type neural network model used in the subsequent analysis:

$$\dot{\mathbf{x}}(t) = -\mathbf{x}(t) + W \tanh(\mathbf{x}(t)), \tag{2}$$

Here, a node denotes one scalar state coordinate of the Hopfield network, rather than an entire chaotic oscillator. The drive and response are two copies of this network. The state vector is $\mathbf{x}(t) = [x_1(t), x_2(t), x_3(t)]^T$. Following the three-neuron Hopfield-type chaotic model studied in [29], in this paper, the network weight matrix is specified as

$$W = \begin{bmatrix} -1.4 & 1.2 & -7.0 \\ 1.1 & 0.0 & 2.8 \\ 0.8 & -2.0 & 4.0 \end{bmatrix}. \tag{3}$$

The network in (2) is taken as the drive system. A single-input pinning controller is introduced into one selected response node, giving

$$\dot{\mathbf{y}}(t) = -\mathbf{y}(t) + W_r \tanh(\mathbf{y}(t)) + Bu(t), \tag{4}$$

where $\mathbf{y}(t) \in \mathbb{R}^3$ is the state of the response system; W_r is the weight matrix of the response system, which may differ from W to represent parameter uncertainty; $u(t) \in \mathbb{R}$ is the scalar control input; and $B \in \mathbb{R}^{3\times 1}$ is the pinning-node selection vector, which determines the specific physical node into which the control signal is injected. Only one of its three components is equal to 1, while the others are 0.

For this parameter setting, the initial condition $\mathbf{x}(0) = (0, 0.01, 0)^{\mathrm{T}}$, which has been used as a representative initial state in previous studies on Hopfield chaotic neural networks [29], is adopted to illustrate the basic chaotic behavior of the system. The fourth-order Runge–Kutta (RK4) trajectories illustrate the reported chaotic regime of this model. To further illustrate the uncontrolled drive–response behavior before applying the proposed pinning controller, two identical Hopfield chaotic neural networks are simulated with the same system parameters but different initial states sampled from Uniform$([-1, 1]^3)$. As shown in Fig. 1(a), the two trajectories evolve around the same double-scroll chaotic attractor but do not naturally synchronize.

The time-domain trajectories of $x_1(t)$, $x_2(t)$, and $x_3(t)$ obtained from the representative initial condition $\mathbf{x}(0) = (0, 0.01, 0)^{\mathrm{T}}$ are shown in Fig. 1(b). The plotted irregular oscillations illustrate the drive dynamics; a finite trajectory plot alone is not a chaos certificate.

Define the synchronization error as $\mathbf{e}(t) = \mathbf{y}(t) - \mathbf{x}(t)$. The ideal objective is vanishing synchronization error [2]. Subtracting the drive dynamics from the response dynamics gives

$$\dot{\mathbf{e}}(t) = -\mathbf{e}(t) + W_r \tanh(\mathbf{y}(t)) - W \tanh(\mathbf{x}(t)) + Bu(t). \tag{5}$$

The following definition mainly corresponds to the nominal synchronization control problem.

Definition 1. For the drive system (2) and the response system with pinning control (4), if under the action of the control law $u(t)$ one has $\lim_{t\to\infty} \|\mathbf{e}(t)\| = 0$, then the systems are said to achieve asymptotic synchronization.

This definition provides the ideal reference. The implemented objective is to reduce synchronization error using a fixed actuator location and bounded input. Here, practical synchronization denotes the observed low-residual behavior over the reported testing windows; it is assessed by full-horizon and terminal error metrics rather than an assumed asymptotic limit. Section 3.3 separately examines boundedness and post-training trajectory sensitivity. Full observation is the default, and reduced policy inputs are evaluated as a separate experimental factor.

3. PPO-Based Single-Input Pinning Control Method

The framework has three stages. Local dynamical and control indicators first identify candidate actuator locations. For each tested location, PPO learns a bounded scalar feedback policy from the nominal simulator. Finally, the frozen policy is evaluated using the implemented sampled dynamics, with state boundedness and local trajectory sensitivity treated separately. The indicators inform the choice of channels to investigate; they are not inputs to the policy or terms in the training reward.

3.1. Local Pinning-Node Selection Analysis

Candidate nodes are compared using complementary local descriptions: transverse growth under ideal coordinate pinning, the regulation cost of a frozen linear model, and a controllability-volume proxy. These descriptions probe different aspects of actuator placement. They support qualitative screening before nonlinear evaluation, without assigning a combined numerical score.

For the local pinning-node analysis, we first consider the baseline identical-system setting, where the drive and response systems share the same weight matrix, i.e., $W_r = W$, and no external disturbance is imposed. When the synchronization error approaches zero, i.e., $\mathbf{e} \to \mathbf{0}$, the nonlinear term can be expanded to the first order around the synchronization manifold. This gives the following local linear approximation of the error system [34]:

$$\dot{\mathbf{e}}(t) \approx [-I + W D(\mathbf{x}(t))]\, \mathbf{e}(t) + Bu(t), \tag{6}$$

where I is the 3×3 identity matrix, W is the weight matrix of the drive system, $u(t) \in \mathbb{R}$ is the scalar control input, and $B \in \mathbb{R}^{3\times1}$ is the control allocation vector indicating the node at which the control signal is injected. The diagonal matrix $D(\mathbf{x})$ is composed of the derivatives of the activation function:

$$D(\mathbf{x}) = \mathrm{diag}\left(1 - \tanh^2(x_1), 1 - \tanh^2(x_2), 1 - \tanh^2(x_3)\right), \tag{7}$$

where x_1, x_2, and x_3 are the state variables of the drive system.

For single-input pinning control, different node selections correspond to different control allocation vectors:

$$B_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad B_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad B_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \tag{8}$$

Here, B_1, B_2, and B_3 represent the cases where the control input is injected into Nodes 1, 2, and 3, respectively.

First, the maximum conditional Lyapunov exponent (MCLE) [35] is used under an ideal coordinate-pinning assumption. If the selected error coordinate is constrained to remain identically zero, let S_i contain the two columns of I_3 associated with the remaining coordinates. Their variational equation is $\dot{\boldsymbol{\zeta}} = S_i^{\mathrm{T}}[-I + W D(\mathbf{x}(t))]S_i\boldsymbol{\zeta}$. This idealized reduced system differs from finite-amplitude, full-error feedback through B_i. Let $\Psi_i(t, 0)$ be the fundamental matrix of this reduced variational system, with $\Psi_i(0, 0) = I_2$. Its largest growth rate is expressed as

$$\lambda_{\max}^{(c)}(B_i) = \limsup_{t\to\infty} \frac{1}{t} \ln \|\Psi_i(t, 0)\|_2, \quad i = 1, 2, 3, \tag{9}$$

where $\|\cdot\|_2$ is the induced Euclidean matrix norm, so the expression accounts for the most amplified perturbation direction. If $\lambda_{\max}^{(c)}(B_i) < 0$, small perturbations in the corresponding transverse subspace tend to decay locally; if $\lambda_{\max}^{(c)}(B_i) > 0$, locally divergent directions still exist in that subspace. In this work, the MCLE is used as a local reference index for node pre-screening under the ideal pinning assumption.

Second, to compare the local control difficulty associated with different pinning nodes, a time-invariant local proxy model is further considered. Specifically, the Jacobian matrix is frozen at the reference point $\mathbf{x} = \mathbf{0}$, where $D(\mathbf{0}) = I$, leading to $A = W - I$. Based on this local approximation, the linear quadratic regulator (LQR) framework [36] is used here as an auxiliary indicator for comparing candidate pinning nodes. An LQR controller derived from the same local model is separately included as a model-based baseline in Section 4.

For each candidate input vector B_i, the standard LQR performance index is introduced as

$$J = \int_0^{\infty} \left(\mathbf{e}^{\mathrm{T}} Q \mathbf{e} + u^{\mathrm{T}} R u\right) dt, \tag{10}$$

where Q is the state weighting matrix and R is the input weighting matrix. The corresponding local state-feedback control law is

$$u = -K_i \mathbf{e}, \tag{11}$$

where the feedback gain matrix is given by

$$K_i = R^{-1} B_i^{\mathrm{T}} P_i, \tag{12}$$

and P_i is determined by the continuous algebraic Riccati equation associated with (A, B_i).

In this paper, the Riccati-based cost index is defined as

$$J_R(B_i) = \mathrm{tr}(P_i), \tag{13}$$

which aggregates the directional costs of the local proxy. For a zero-mean initial error with identity covariance, the expected optimal linear cost is $\mathbb{E}[\mathbf{e}(0)^{\mathrm{T}} P_i \mathbf{e}(0)] = \mathrm{tr}(P_i)$. Thus the index compares nodes under a common state scaling and weighting, while $\|K_i\|$ describes local feedback strength. Neither quantity is the realized cost of the saturated nonlinear policy.

Finally, to measure how effectively the control action propagates from different nodes to the entire local state space, the same frozen local matrix $A = W - I$ is used.

Based on this matrix, the controllability matrices corresponding to the three candidate pinning nodes are constructed as [37]

$$C_i = \left[B_i,\ AB_i,\ A^2 B_i\right], \quad i = 1, 2, 3, \tag{14}$$

and the local control volume index is defined as

$$V_i = \det(C_i)^2. \tag{15}$$

Here, C_i characterizes the ability of the control input injected through the ith node to propagate over the entire local state space. Since this determinant-based index depends on the chosen state scaling, it is used only as a relative local propagation proxy among candidate nodes under the same coordinate system.

Table 1
Local auxiliary indicators for candidate screening.

Pinning Node	$\lambda^{(c)}_{\max}$	J_R	$\|K_i\|$	V_i
Node 1	+0.5858	280.2686	20.3937	0.479
Node 2	+1.5892	15.5170	6.7286	262.18
Node 3	-0.5890	7.8327	4.3829	290.77

For the MCLE calculation, the drive initial state is

$$\mathbf{x}(0) = (0.5, -0.01, 0.2)^{\mathrm{T}}.$$

The drive is integrated for 1000 time units, with the first 50 discarded. The state and reduced variational equations use RK4 with QR renormalization at each step. Step sizes 0.01, 0.005, and 0.0025 give the following ranges:

$$\text{Node 1: } [0.5858, 0.5982],$$
$$\text{Node 2: } [1.5854, 1.5900],$$
$$\text{Node 3: } [-0.5890, -0.5841].$$

Their signs are unchanged. The finest-step estimates replace the earlier mixed RK4/Euler estimates, but the displayed digits are not certified infinite-time values.

Table 1 shows that Node 3 has a negative ideal-pinning MCLE estimate, the smallest J_R and $\|K_i\|$, and the largest V_i. These concordant local indicators motivate its selection as a preferred candidate in this system, subject to nonlinear closed-loop evaluation.

Node 2 has a larger controllability-volume proxy and a lower Riccati cost than Node 1, although its ideal-pinning transverse growth estimate is higher. The indices therefore do not yield an identical ordering of these two channels. Node 1 has the largest Riccati cost and gain norm and the smallest volume proxy; these properties motivate testing it as a less favorable candidate under the common bounded-input learning setup.

The MCLE alone does not order Nodes 1 and 2 in the same way as the other indicators. Node 2 is retained ahead of Node 1 because its Riccati cost and gain norm are substantially smaller and its local volume is larger; no weighted aggregate score or general ranking rule is proposed. Using $\succ$ to denote screening preference, the resulting qualitative order for this system is

$$\text{Node 3} \succ \text{Node 2} \succ \text{Node 1}. \tag{16}$$

The resulting experimental design retains all three single-node policies for nominal comparison. Node 3 supplies the favorable reference channel, and Node 2 supplies a contrasting channel for the subsequent observation and robustness assessments.

All three frozen controllability matrices have rank three. Accordingly, a positive ideal-pinning MCLE does not imply uncontrollability of (A, B_i) or rule out synchronization by a designed traditional feedback law. The designation "challenging" describes relative behavior in this model and the tested constrained controllers, not impossibility of linear control at Node 2.

The ideal-pinning exponent follows the drive trajectory, whereas the Riccati and volume indices freeze the Jacobian at the origin. Their agreement for Node 3 is useful screening evidence, but a broader relation between these proxies and learned performance would require additional networks and parameter settings. Section 4 checks their relevance to the present nonlinear system.

3.2. PPO-Based Control Formulation and Training Procedure

With fixed nominal parameters, the sampled drive–response state defines a continuous-state, continuous-action Markov decision process. The full observation $(\mathbf{x}, \mathbf{e})$ determines both system states because $\mathbf{y} = \mathbf{x} + \mathbf{e}$. A policy is trained separately for each actuator location; it does not switch the controlled node during an episode. Reduced observations are treated separately in Section 4.2.

To enable the agent to perceive both the instantaneous dynamical state of the drive system and the current synchronization deviation, the full observation vector is defined as

$$O_t = \frac{1}{c_o}\left[x_1(t), x_2(t), x_3(t), e_1(t), e_2(t), e_3(t)\right]^{\mathrm{T}}, \tag{17}$$

where $c_o = 5.0$ is the fixed observation scaling used in implementation. The notation c_o is used here to avoid confusion with the Gaussian noise intensity σ used in the robustness experiments.

For PPO, the sampled Gaussian policy action during training, or its deterministic mean during evaluation, is clipped to $[-1, 1]$ before deployment. Denoting this clipped action by a_t, it is mapped to the physical control input as

$$u(t) = \kappa a_t, \tag{18}$$

where κ is the control scaling factor, and $\kappa = 4$ is adopted in this paper.

Let $t_k = k\Delta t$ denote the control-update times. The applied input $u_k = \kappa a_k$ is held constant on $[t_k, t_{k+1})$.

The environment then advances the drive and response and returns a reward based on the updated error:

$$r_k = -0.1\|\mathbf{e}(t_{k+1})\|_1 = -0.1\sum_{i=1}^{3}|e_i(t_{k+1})|. \tag{19}$$

Maximizing the discounted reward promotes small synchronization errors throughout the rollout. The reward contains no control-effort penalty; input magnitude is constrained by action clipping, and effort is measured separately during evaluation. The same error reward is retained in the reduced-observation and multi-input experiments to keep their learning objectives consistent.

PPO is selected for its direct Gaussian-policy parameterization and alternating rollout and minibatch-update implementation [38]. These features fit the sampled scalar-feedback task. The choice is evaluated empirically against SAC in Section 4.4; it does not require PPO to be the best learning algorithm for every operating condition.

To learn a control policy in the above continuous-action setting, this paper employs the PPO algorithm [38]. PPO is an Actor–Critic method whose clipped surrogate discourages excessively large policy changes; this does not enforce a hard bound on policy updates or guarantee sample efficiency. This likelihood-ratio clipping is distinct from actuator clipping. The PPO objective function is written as

$$L(\theta) = \hat{\mathbb{E}}_t\left[\min\left(\rho_t(\theta)\hat{A}_t,\ \text{clip}\left(\rho_t(\theta), 1-\epsilon, 1+\epsilon\right)\hat{A}_t\right)\right], \tag{20}$$

where

$$\rho_t(\theta) = \frac{\pi_\theta(\widetilde{a}_t|O_t)}{\pi_{\theta_{\text{old}}}(\widetilde{a}_t|O_t)}. \tag{21}$$

Here $\widetilde{a}_t$ is the unclipped Gaussian sample stored for the policy update, whereas $a_t = \text{clip}(\widetilde{a}_t, -1, 1)$ is applied to the environment. The ratio $\rho_t(\theta)$ compares the new and old action likelihoods, ϵ is the PPO clipping hyperparameter, and $\hat{A}_t$ is obtained using generalized advantage estimation (GAE). Optimization aims to reduce the discounted synchronization-error cost; attainment of its global minimum is not asserted.

As shown in Fig. 2, PPO follows an interaction–update training process. The agent first interacts with the environment using the current policy and stores the collected states, actions, rewards, and value estimates in a trajectory buffer. The advantage function is then estimated, typically using GAE, and the policy network is updated by maximizing a clipped surrogate objective. Meanwhile, the value network is trained by minimizing the mean squared error between the predicted and target state values. The clipped surrogate discourages some large likelihood-ratio changes during optimization. It does not impose a hard bound on the change in the deployed feedback law or certify closed-loop stability.

The overall training procedure is summarized in Algorithm 1. In each iteration, trajectories are first collected by executing the current policy in the error-system environment. The collected data are then used to estimate the advantage function and update the policy network with the PPO clipped objective. The value network is updated by minimizing the squared value loss. After repeated interaction and optimization, the trained policy is used as the single-input pinning controller for the response system.

3.3. Boundedness and Post-Training Local Numerical Assessment

Before evaluating synchronization performance, it is useful to distinguish boundedness of the controlled trajectories, sensitivity of a deployed closed-loop trajectory to small error perturbations, and convergence of the PPO training algorithm. Boundedness follows from the dissipative Hopfield dynamics and the bounded actuator. Local sensitivity is assessed numerically after training using the closed-loop control-step Jacobian. Neither convergence of PPO training nor a formal synchronization stability guarantee is established here.

Because the normalized action satisfies $a_t \in [-1, 1]$ and the physical action is defined by (18), the deployed control input satisfies

$$|u(t)| \le \kappa, \qquad \kappa = 4. \tag{22}$$

Moreover, each single-node pinning vector has $\|B_i\|_2 = 1$.

Proposition 3.1 (Boundedness of the controlled drive–response trajectories)**.** *Consider the drive system (2) and response system (4), where W and W_r are finite constant matrices and the control input is measurable and satisfies (22). This includes the deployed piecewise-constant sample-and-hold input. For every finite initial condition, the corresponding trajectories are forward complete and uniformly ultimately bounded. In particular,*

$$\limsup_{t\to\infty}\|\mathbf{x}(t)\|_2 \le \sqrt{3}\|W\|_2, \tag{23}$$

$$\limsup_{t\to\infty}\|\mathbf{y}(t)\|_2 \le \sqrt{3}\|W_r\|_2 + \kappa. \tag{24}$$

Proof. Since $\|\tanh(\mathbf{z})\|_2 \le \sqrt{3}$ for every $\mathbf{z} \in \mathbb{R}^3$, the Cauchy–Schwarz inequality and the induced matrix norm give

$$\begin{aligned} D^+\|\mathbf{x}\|_2 &\le -\|\mathbf{x}\|_2 + \sqrt{3}\|W\|_2, \\ D^+\|\mathbf{y}\|_2 &\le -\|\mathbf{y}\|_2 + \sqrt{3}\|W_r\|_2 + \|B_i\|_2|u(t)| \\ &\le -\|\mathbf{y}\|_2 + \sqrt{3}\|W_r\|_2 + \kappa, \end{aligned} \tag{25}$$

where D^+ is the upper right Dini derivative, $\|B_i\|_2 = 1$, and (22) has been used. Applying the scalar comparison lemma to (25) yields

$$\|\mathbf{x}(t)\|_2 \le e^{-t}\|\mathbf{x}(0)\|_2 + \sqrt{3}\|W\|_2(1-e^{-t}). \tag{26}$$

$$\|\mathbf{y}(t)\|_2 \le e^{-t}\|\mathbf{y}(0)\|_2 + \left(\sqrt{3}\|W_r\|_2 + \kappa\right)(1-e^{-t}). \tag{27}$$

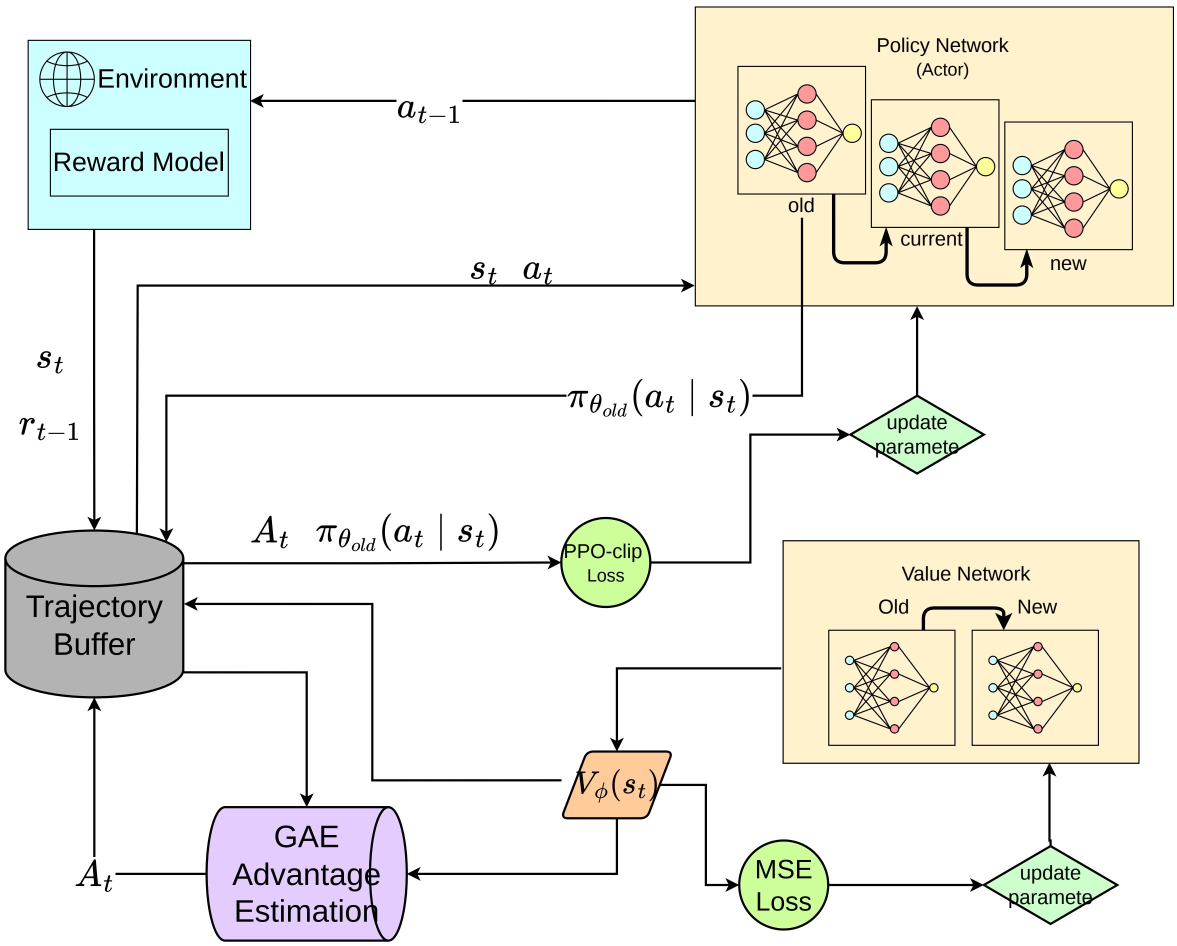


Figure 2: General training process of the PPO-based DRL framework.

Taking the upper limit as $t \to \infty$ gives (23) and (24). For a prescribed bounded measurable input, the vector field is globally Lipschitz in the state and has bounded forcing. The explicit bounds preclude finite escape; the sampled feedback can consequently be extended over successive control intervals for all future times. □

For the nominal matched case $W_r = W$, the matrix in (3) has $\|W\|_2 = 8.9524$. Proposition 3.1 therefore gives the conservative bounds $\limsup_{t\to\infty} \|\mathbf{x}(t)\|_2 \leq 15.5059$ and $\limsup_{t\to\infty} \|\mathbf{y}(t)\|_2 \leq 19.5059$. These bounds are not intended to predict the observed attractor size. They establish non-divergence using worst-case norm inequalities. In particular, boundedness of both trajectories alone does not imply that the synchronization error converges to zero.

The MCLE $\lambda^{(c)}_{\max}(B_i)$ in Section 3.1 and the closed-loop exponent introduced below have different purposes. The former is a pre-training node-screening indicator derived from a locally linearized transverse model under the ideal pinning assumption and does not include the learned PPO policy. By contrast, the latter is evaluated after training from the actual deployed sample-and-hold closed-loop map. Consequently, their numerical values should not be directly compared or interpreted as equivalent stability measures.

The implementation uses a sample-and-hold controller: one action is evaluated every five RK4 substeps and is held constant over the control interval $\Delta t = 0.05$. Accordingly, local behavior of the trained controller is assessed using the actual discrete control-step map

$$\mathbf{x}_{k+1} = \Phi_x(\mathbf{x}_k), \qquad \mathbf{e}_{k+1} = G_i(\mathbf{x}_k, \mathbf{e}_k; \pi_{\theta,i}), \tag{28}$$

where G_i includes the deterministic mean action of the trained policy, action clipping, the physical scale κ, and the five RK4 substeps. The transverse Jacobian of the deployed map is

$$M_{i,k} = \frac{\partial G_i}{\partial \mathbf{e}}(\mathbf{x}_k, \mathbf{e}_k; \pi_{\theta,i}). \tag{29}$$

The asymptotic largest conditional Lyapunov exponent is defined by

$$\lambda_i^{\mathrm{cl}} = \limsup_{N\to\infty} \frac{1}{N\Delta t} \ln \left\| M_{i,N-1} M_{i,N-2} \cdots M_{i,0} \right\|_2. \tag{30}$$

In practice, automatic differentiation and QR renormalization yield a finite-time estimate $\widehat{\lambda}_i^{\mathrm{cl}}$ over the stated observation window, not an evaluation of the infinite-time limit. A negative estimate describes average decay of linearized perturbations about the evaluated response trajectory while holding the drive trajectory fixed. It does not establish convergence of the synchronization error itself: the reference trajectory can have a nonzero residual. No uniform contraction region, certified attraction basin, or small ultimate bound on synchronization error is established. Parameter mismatch, impulse disturbance, and Gaussian noise are therefore separate empirical robustness tests in Section 4, not consequences of Proposition 3.1 or (30).

The numerical assessment was conducted for Nodes 2 and 3 using three independently trained policies for each node. Each policy was evaluated from ten initial conditions, giving 30 closed-loop trajectories per node and 60 trajectories in total. Each trajectory contained 2000 control intervals, and the first 100 intervals were discarded as a transient, leaving 95 time units for exponent accumulation. Table 2 summarizes the largest conditional Lyapunov exponent and the post-transient synchronization residual. The reported mean error is obtained by first averaging $\|\mathbf{e}_k\|_2$ over the post-transient portion of each trajectory and then averaging over the 30 trajectories for the corresponding node.

These trajectories use three independently trained policies and ten initial conditions per policy. Replaying all 60 trajectories reproduced the reported mean residuals exactly. No state-guard activation occurred at any RK4 substep; the largest absolute state component was 3.9134, below the implementation guard at 10. Consequently, omission of that inactive guard from the differentiated nominal map does not change its local derivative along these trajectories. At action-clipping thresholds the map is nonsmooth, so automatic derivatives are understood away from those thresholds, not as a global differentiability assertion.

Table 2: Finite-time conditional Lyapunov estimates for the deployed sample-and-hold PPO controllers.

Node	$\widehat{\lambda}_i^{\mathrm{cl}}$	Range	Mean $\|\mathbf{e}\|_2$
2	−1.117	[−1.317, −0.872]	0.0276
3	−0.581	[−0.592, −0.557]	0.0075

The finite-time estimate is negative for all 60 evaluated trajectories. A shorter check using 600 control intervals produced the same signs and similar aggregate values, supporting numerical consistency over the examined windows, not convergence to an infinite-time value. Node 2 has a more negative mean estimate, whereas Node 3 has a smaller post-transient error. This is not contradictory: local perturbation decay and residual synchronization error are different quantities. In particular, the policy architecture does not impose $\pi_{\theta,i}(\mathbf{x},\mathbf{0}) = 0$. Thus, the synchronization manifold need not be invariant. Here "practical synchronization" describes the observed low-residual behavior; it does not denote a proved ultimate-error bound or asymptotic synchronization theorem.

As a complementary multi-step check, a positive-definite metric $\boldsymbol{P}$ was fitted once per policy using a separate nominal rollout. In the induced norm $\|\mathbf{v}\|_P = (\mathbf{v}^\top \boldsymbol{P}\mathbf{v})^{1/2}$, the largest sampled gains of the ordered 40-step Jacobian products were 0.4882 and 0.5340 for Nodes 2 and 3, respectively; at 80 steps they were 0.0673 and 0.2802. These checks used 60 nominal trajectories of 2,000 control intervals, excluding the first 100 intervals. Forty control intervals correspond to two system time units. Direct paired-rollout checks with initial response perturbations up to 0.05 along signed coordinate directions also showed contraction at these horizons.

These results strengthen the evidence for local perturbation attenuation over finite windows, even when individual steps can amplify perturbations. They do not establish contraction throughout a continuous neighborhood, a synchronization-error bound, or robustness to parameter mismatch. In particular, convergence toward a reference response trajectory is distinct from exact agreement with the drive trajectory, which must be assessed separately for the discrete application.

4. Experimental Results and Analysis

The experiments address three questions: how actuator location affects PPO performance and observation demand; what changes when additional actuators become available; and how the frozen PPO policies compare with alternative controllers under parameter mismatch [39], impulse disturbances, and process noise. The comparisons report synchronization error and applied control effort under the stated shared test conditions.

4.1. Experimental Setup and Evaluation Metrics

The programs were implemented in Python using Stable Baselines3. The workstation is equipped with an NVIDIA RTX 5070Ti GPU; the policy training scripts considered here use the CPU. PPO used a nominal training budget of 4×10^5 environment transitions, with four parallel environments and 512 steps per environment in each rollout.

During environment interaction, the continuous-time system dynamics were integrated using the classical fourth-order Runge–Kutta (RK4) method. The basic integration step size was set to 0.01 time units. To improve numerical stability, five RK4 substeps were performed within each control step. Therefore, each control update corresponded to an effective time interval of 0.05 time units. The maximum magnitude of the physical control input was constrained to 4 units.

The implementation also contains a numerical state guard at $[-10, 10]$ after RK4 substeps and disturbance updates. No guard activation was detected in the 2,940 reevaluated episodes covering the traditional baselines, matched-budget SAC comparison, and actuator-count ablation; the reported trajectories in these comparisons therefore evolve without state truncation.

At initialization, the initial states of both the drive system and the response system, denoted by $\mathbf{x}(0)$ and $\mathbf{y}(0)$, were sampled from the uniform distribution $[-1.0, 1.0]$. This common distribution provides varied initial synchronization errors and is held fixed across the compared methods. The reported results are conditional on this initialization range.

Algorithm 1 PPO-based Single-Input Pinning Control for the Error System

Require: Error-system environment $\mathcal{E}$; policy network π_θ; value network V_ϕ; learning rate η; clipping parameter ϵ; discount factor γ; GAE parameter λ.
Ensure: Optimized policy parameters θ.

Initialize policy parameters θ_0 and value-function parameters ϕ_0.
Initialize the drive–response system and obtain the initial observation O_0.
Define a trajectory buffer Γ to store observations, actions, rewards, and value estimates.
for $k = 0, 1, 2, \dots, K-1$ **do**
  Collect trajectories Γ_k by executing the current policy π_{θ_k} in the environment.
  for each time step t in Γ_k **do**
    Sample $\widetilde{a}_t \sim \pi_{\theta_k}(\cdot|O_t)$; set $a_t = \mathrm{clip}(\widetilde{a}_t, -1, 1)$.
    Store the unclipped sample for the policy update; apply $u(t) = \kappa a_t$.
    Apply $u(t)$ to the response system and observe reward r_t and next observation O_{t+1}.
  end for
  Compute advantage estimates $\hat{A}_t$ using generalized advantage estimation.
  Update policy parameters by maximizing the PPO clipped objective:

$$\theta_{k+1} = \arg\max_\theta \hat{\mathbb{E}}_t \Big[\min\big(\rho_t(\theta)\hat{A}_t, \mathrm{clip}(\rho_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t\big)\Big].$$

  where

$$\rho_t(\theta) = \frac{\pi_\theta(\widetilde{a}_t|O_t)}{\pi_{\theta_k}(\widetilde{a}_t|O_t)}.$$

  Update value-function parameters by minimizing the squared value loss:

$$\phi_{k+1} = \arg\min_\phi \hat{\mathbb{E}}_t \left[\left(V_\phi(O_t) - \hat{R}_t\right)^2 \right].$$

end for
return trained policy π_θ.

For the reinforcement learning configuration, both the policy network and the value network were represented by multilayer perceptrons (MLPs). The Adam optimizer was adopted with a learning rate of 3×10^{-4}, so as to balance exploration efficiency and policy-update stability.

Three traditional controllers are included to provide references of increasing model dependence. First, the original fixed-gain feedback baseline [5] is retained without retuning:

$$u(t) = \begin{cases} -4, & -ke_i(t) < -4, \\ -ke_i(t), & -4 \le -ke_i(t) \le 4, \\ 4, & -ke_i(t) > 4, \end{cases} \tag{31}$$

where $e_i(t)$ denotes the synchronization error at the controlled node i. This baseline represents a simple local-error controller rather than an optimally tuned linear design. The previously selected gain $k = 8.0$ is used throughout all reported experiments.

Second, a continuous-time LQR baseline is obtained from the frozen local error model in Section 3. With $Q = I_3$ and $R = 1$, its saturated control law is

$$u_{\mathrm{LQR},i}(t) = \mathrm{sat}_4\left[-K_i \mathbf{e}(t)\right], \tag{32}$$

where $K_i = R^{-1}B_i^{\mathrm{T}}P_i$ and P_i is the stabilizing solution of the continuous algebraic Riccati equation associated with (A, B_i). The resulting gains are

$$K_2 = (-0.116727,\ 3.871191,\ -5.502161),$$
$$K_3 = (-0.171698,\ -2.085641,\ 3.850986).$$

Unlike fixed-gain feedback, LQR uses all error components and the nominal local model. Unless stated otherwise, the traditional-controller figures use this reference design with $R = 1$. Section 4.3.4 reports a separately validated weight-sensitivity comparison. Optimality for the saturated nonlinear system is not claimed.

Third, a boundary-layer sliding-mode controller (SMC) is included as a nonlinear model-based baseline [7, 8]. Define

$$\mathbf{g}_0(\mathbf{x}, \mathbf{y}) = -\mathbf{e} + W\left[\tanh(\mathbf{y}) - \tanh(\mathbf{x})\right], \tag{33}$$

and select $c_i^{\mathrm{T}} = K_i/(K_iB_i)$ so that $c_i^{\mathrm{T}}B_i = 1$. The sliding variable and control law are

$$s_i = c_i^{\mathrm{T}}\mathbf{e}, \qquad u_{\mathrm{SMC},i} = \mathrm{sat}_4\left[-c_i^{\mathrm{T}}\mathbf{g}_0 - k_s\,\mathrm{sat}\left(\frac{s_i}{\phi}\right)\right], \tag{34}$$

where $\mathrm{sat}(z) = \max(-1, \min(1, z))$, $k_s = 16$, and $\phi = 0.5$. These parameters minimize the recorded validation score over $k_s \in \{4, 8, 12, 16\}$ and $\phi \in \{0.2, 0.5, 1, 2\}$, using ten validation initial conditions separate from the test seeds. Validation includes nominal, mismatch ($\eta = 1$), and impulse scenarios; the score averages their MAEs normalized by 0.1, 0.15, and 0.45, respectively. Parameters are then frozen. Thus, although its compensation term uses the nominal model, SMC parameter selection has access to non-ideal validation scenarios. The saturated, sampled implementation is evaluated empirically; the continuous-time ideal sliding condition is not asserted to hold when the actuator saturates.

All four methods use the same RK4 solver, control-update interval, initial-state distribution, testing horizon (400 updates, or 20 time units), and actuator bound $|u| \le 4$. The PPO results aggregate three independently trained policies, each evaluated on 20 common test seeds ; each deterministic traditional controller is evaluated on the same 20 test seeds. For each scenario, the initial conditions and exogenous random realizations are shared across controllers. In the existing environment, drawing a mismatch matrix consumes random numbers before initialization; hence a fixed seed does not imply identical initial states between $\eta = 0$ and $\eta > 0$. Comparisons between controllers are paired within each scenario.

PPO is trained only on the nominal system. LQR uses $\mathbf{e}$ and the nominal local model, while SMC additionally uses $\mathbf{x}$ and $\mathbf{y}$ for nonlinear compensation. PPO receives $(\mathbf{x}, \mathbf{e})/5$ without evaluating an analytical model during deployment; its training simulator nevertheless uses the nominal dynamics. Fixed-gain feedback uses only e_i. Consequently, these are comparisons under matched actuation constraints, not identical information or design objectives. In particular, the PPO reward penalizes synchronization error but not control energy, whereas LQR explicitly penalizes both. Lower error at higher energy is therefore interpreted as a performance–effort tradeoff, not evidence of superior energy efficiency.

To evaluate the synchronization performance of different control methods, several metrics are introduced from the perspectives of overall synchronization accuracy, steady-state error, transient recovery capability, and control effort. Let $T = 400$ denote the number of control updates and $N = 3$ the system dimension. For the baseline and ablation comparisons, errors are recorded at the $T + 1$ sampling instants including initialization. The synchronization error vector is

$$\mathbf{e}(t) = \mathbf{y}(t) - \mathbf{x}(t) = [e_1(t), e_2(t), \dots, e_N(t)]^{\mathrm{T}}. \tag{35}$$

The mean absolute error (MAE) is used to measure the average synchronization deviation over the whole testing interval:

$$\mathrm{MAE} = \frac{1}{(T+1)N} \sum_{t=0}^{T} \sum_{i=1}^{N} |e_i(t)|. \tag{36}$$

MAE provides an intuitive measure of the overall error level by assigning equal weights to different time steps and state components.

The root mean square error (RMSE) is used to characterize the overall error level while being more sensitive to large deviations:

$$\mathrm{RMSE} = \sqrt{\frac{1}{(T+1)N} \sum_{t=0}^{T} \sum_{i=1}^{N} e_i^2(t)}. \tag{37}$$

Compared with MAE, RMSE is more sensitive to transient large errors and is therefore useful for evaluating error fluctuations under parameter mismatch and impulse disturbance.

The final L1 error is used to measure the synchronization accuracy at the end of testing:

$$E_f = \|\mathbf{e}(T)\|_1 = \sum_{i=1}^{N} |e_i(T)|. \tag{38}$$

This metric reflects how closely the response system approaches the drive system at the final testing instant.

In the impulse disturbance experiment, the maximum post-pulse error is used to describe the largest error overshoot after the disturbance is injected:

$$M_p = \max_{t \ge t_p} \|\mathbf{e}(t)\|_1, \tag{39}$$

where t_p denotes the impulse injection step. The imposed state jump largely determines the immediate peak, so M_p alone is not interpreted as a controller-dependent recovery measure.

The post-pulse mean error is defined as the average global synchronization error from the impulse injection time to the end of testing:

$$E_{\mathrm{post}} = \frac{1}{T - t_p + 1} \sum_{t=t_p}^{T} \|\mathbf{e}(t)\|_1, \tag{40}$$

where t_p is the impulse injection time and T is the final testing step.

The control energy is used to measure the cumulative control effort over the testing process, summing over all m active input channels:

$$E_u = \sum_{t=1}^{T} \sum_{j=1}^{m} u_j^2(t) \Delta t, \tag{41}$$

where $\Delta t = 0.05$ is the effective control update interval and $u_j(t)$ is the applied, saturated input during update t. The single-input comparisons have $m = 1$. The steady L1 error averages $\|\mathbf{e}(t)\|_1$ over samples $t = 320, \dots, 400$ (the final four time units), and differs from the terminal error E_f. Reported means first average the test initial conditions for each policy and then average policy seeds; the deterministic baselines have only the first level. Repeated initial conditions for the same trained policy are not treated as independent training replications.

In summary, MAE and RMSE are used to evaluate the overall synchronization performance, Final L1 Error is used to describe the terminal synchronization accuracy, M_p and E_{post} are used to characterize transient recovery capability under impulse disturbance, and E_u is used to assess the control cost.

4.2. Pinning-Node Selection and Observation Demand

To evaluate the guidance provided by the local node-selection analysis in Section 3, the single control input was applied to Nodes 1, 2, and 3 under the same PPO framework and training configuration. This experiment compares achieved synchronization performance under a fixed learning budget; it is not a controllability or reachability test.

As shown in Fig. 3, Node 3 has faster observed error decay and a lower residual than Node 2. Node 1 exhibits the weakest learned control performance under the same training budget. These differences describe the trained controllers

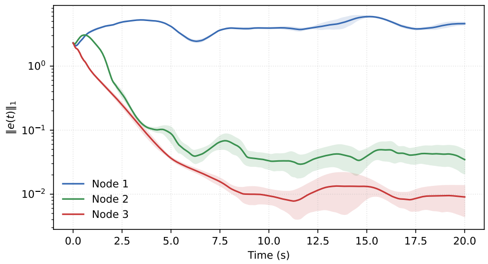


Figure 3: Synchronization error under different pinning nodes. Each curve averages three training-seed means, each evaluated on 20 common initial conditions; shading shows one sample standard deviation across the three seed means.

and do not establish intrinsic limits on what each actuator location can achieve.

To avoid relying only on a single error curve for qualitative judgment, MAE, RMSE, and Final L1 Error are further used to quantitatively compare the synchronization performance under different pinning nodes. The results in Table 3 further confirm the above observation. In terms of both average error and terminal error, Node 3 achieves the best overall synchronization performance. This result is generally consistent with the local auxiliary indicators in Table 1, including the $\lambda_{\max}^{(c)}$, J_R, $\|K_i\|$, and V_i. Thus the favorable Node 3 screening result is reflected in the tested policies. Agreement within this network supports the use of the indicators as local diagnostics; it does not establish a general predictor of PPO performance.

Table 3
Quantitative synchronization performance under different pinning nodes, using three training seeds and 20 common test initial conditions per seed. MAE and RMSE are computed from the error components, consistently with the baseline and actuator-count comparisons.

Node	MAE	RMSE	Final L1 Error E_f
Node 1	1.3937	1.6342	4.6052
Node 2	0.0930	0.2863	0.0347
Node 3	0.0431	0.1434	0.0091

To show component-level behavior, Fig. 4 restores the original three-dimensional error surfaces for multiple initial conditions. Surface height represents the signed error component, with time and test index on the other axes. The displayed trajectories illustrate decay toward low error at Nodes 3 and 2 and persistent fluctuations at Node 1. They provide a qualitative illustration, not additional independent training replications; quantitative comparisons use Table 3. Test index is categorical, so the surfaces between trajectories are a visual interpolation rather than a physical state dimension.

Node 3 is retained as the favorable candidate and Node 2 as a contrasting, more challenging learned-control channel. Both are evaluated in the subsequent robustness and algorithm comparisons, allowing actuator placement to be considered alongside controller choice.

4.2.1. *Partial-observation assessment*

Partial observability is next considered to examine how the information demand of the learned policy depends on the selected pinning node [20]. Within each node, the actuation structure and training procedure are retained while a separate policy is trained for each observation configuration. Thus, this comparison concerns the performance attainable by the trained configurations, not the effect of removing measurements from one unchanged policy.

For Node 3, the full-observation policy has access to all drive states and synchronization errors. The four-dimensional and two-dimensional observations are

$$O_{4D} = \{x_1, x_3, e_1, e_3\}, \tag{42}$$

and

$$O_{2D} = \{x_3, e_3\}, \tag{43}$$

respectively. For Node 2, the corresponding observations are $\{x_1, x_2, e_1, e_2\}$ and $\{x_2, e_2\}$.

The four-dimensional subsets retain the controlled coordinate and coordinate 1 as a fixed intermediate configuration; they are not claimed to be optimal sensor subsets. Reduced observation refers to the policy input at deployment. The simulator still computes the training reward from all three error components in (19), so these experiments do not demonstrate training using only local measurements. A memoryless policy is used with the reduced observation, which need not itself be Markovian; recurrent policies and exhaustive sensor-subset comparisons are outside this assessment.

Fig. 5 and Table 4 show similar later-stage errors for the Node 3 full- and four-dimensional policies. The tested two-dimensional policies have higher MAE and terminal error. This is an empirical association with observation reduction, not a separate identification of observability or of the policy's internal inference mechanism. To compare the observation penalty within each node, normalize MAE by that node's six-dimensional value. The four- and two-dimensional penalties are approximately 10.5 and 43.7 at Node 2, versus 1.0 and 1.3 at Node 3. These descriptive ratios, calculated from the common-seed means in Table 4, quantify the actuator-dependent consequence of reducing the selected policy inputs; they are not sensor-optimality or statistical-significance claims.

With these particular subsets, Node 2 shows a larger degradation than Node 3. However, dimension reduction also changes coordinate composition, so these results alone cannot attribute the degradation to observation dimension. The following matched-dimension comparison examines this distinction.

Observation composition at fixed dimension. To separate coordinate composition from input dimension, all three paired subsets $\mathcal{O}_{ij} = \{x_i, x_j, e_i, e_j\}$, with $ij \in \{12, 13, 23\}$,

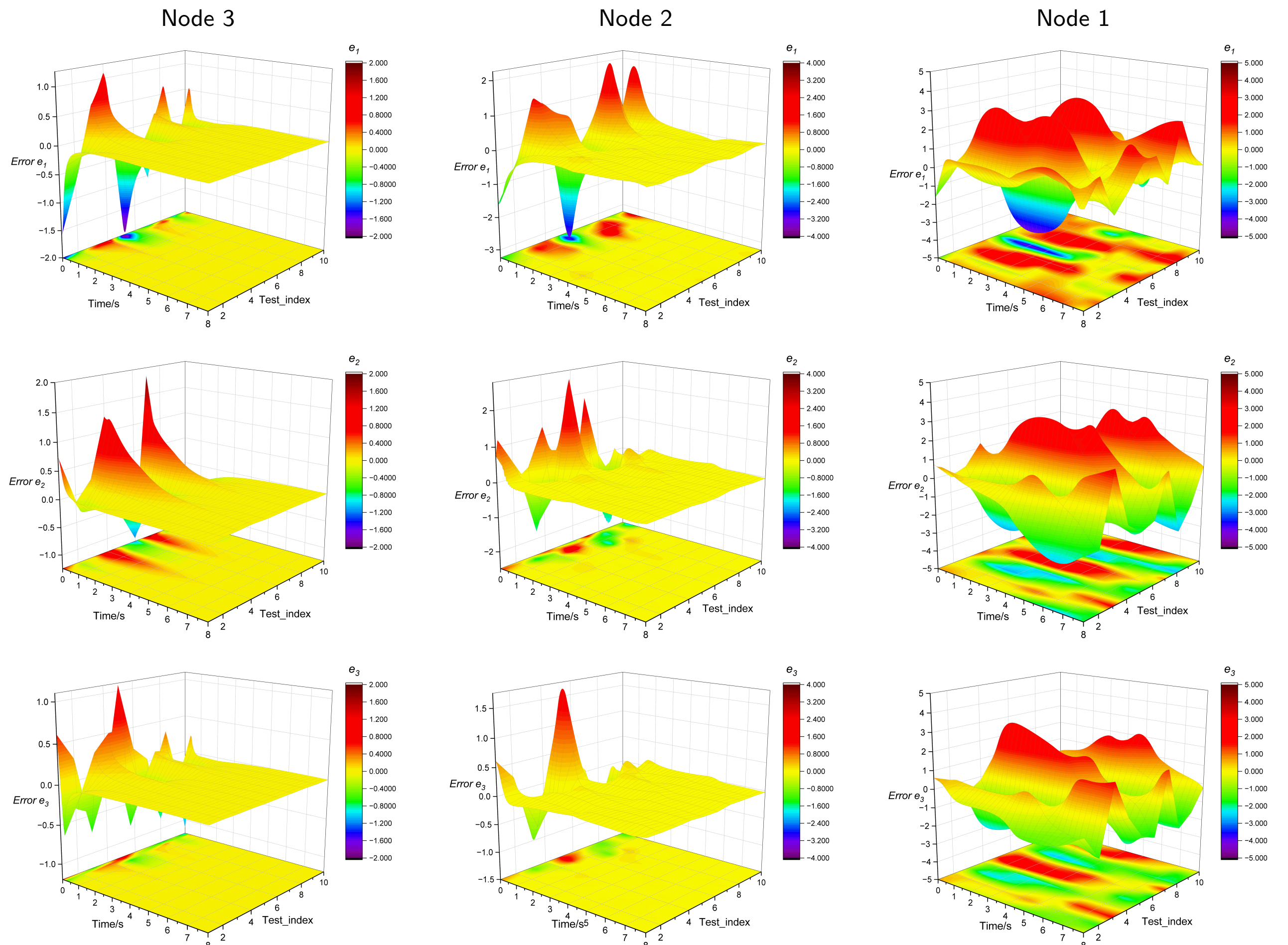


Figure 4: Original three-dimensional synchronization-error surfaces for multiple initial conditions. Columns correspond to actuation at Nodes 3, 2, and 1; within each column, panels show e_1, e_2, and e_3 from top to bottom. Height and color represent signed error. The original axis and color ranges are retained and differ between panels; colors alone should not be used for quantitative comparisons across nodes.

Table 4
Synchronization performance under different observation settings, restricted to the same three training seeds and ten common test initial conditions. MAE is obtained by dividing the recorded time-averaged L1 error by $N = 3$. Observation components are specified in Eqs. 42–43.

Observation Setting	MAE	Final L1 Error
Node 3, 6D	0.0398	0.0081
Node 3, 4D	0.0390	0.0055
Node 3, 2D	0.0524	0.0487
Node 2, 6D	0.0761	0.0198
Node 2, 4D	0.7974	2.4874
Node 2, 2D	3.3271	11.3096

are evaluated at each of Nodes 2 and 3. This completes the two-actuator-location by three-paired-subset comparison, not all arbitrary four-coordinate subsets. Each configuration has three independently trained policies with a nominal budget of 4×10^5 transitions. Six additional policies complete the previously missing Node 2/$\mathcal{O}_{13}$ and Node 3/$\mathcal{O}_{12}$ configurations. All six configurations are evaluated on 20 new common initial conditions, with unchanged dynamics, reward, action bounds, and PPO settings. Every final checkpoint is retained. This evaluation set differs from that in the preceding dimension-reduction comparison.

Table 5
Complete paired four-dimensional observation comparison at Nodes 2 and 3. Entries are mean ± sample standard deviation across three training-run means over 20 common initial conditions.

Node	Subset	MAE	E_u
2	$\mathcal{O}_{12}$	0.9104 ± 0.1312	9.60 ± 3.59
2	$\mathcal{O}_{13}$	0.6253 ± 0.0032	150.68 ± 38.64
2	$\mathcal{O}_{23}$	0.0866 ± 0.0059	9.64 ± 0.81
3	$\mathcal{O}_{12}$	0.5235 ± 0.0115	209.02 ± 25.40
3	$\mathcal{O}_{13}$	0.0438 ± 0.0009	6.85 ± 0.39
3	$\mathcal{O}_{23}$	0.0771 ± 0.0175	142.70 ± 154.59

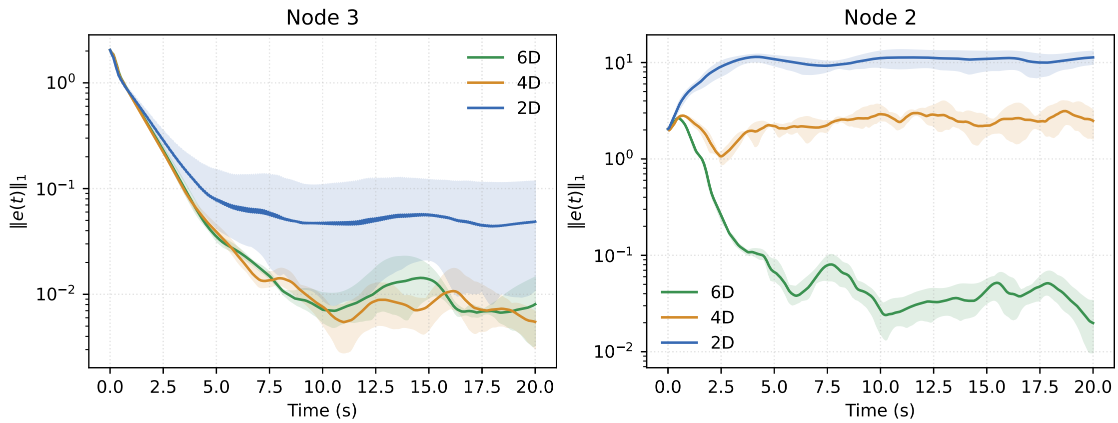


Figure 5: Comparison of L1 synchronization error under different observation configurations. Curves average three training-seed means, with ten common archived test initial conditions per seed; shading spans the minimum and maximum of the three seed means. This archived observation experiment uses fewer test initial conditions than the baseline and actuator-count comparisons.

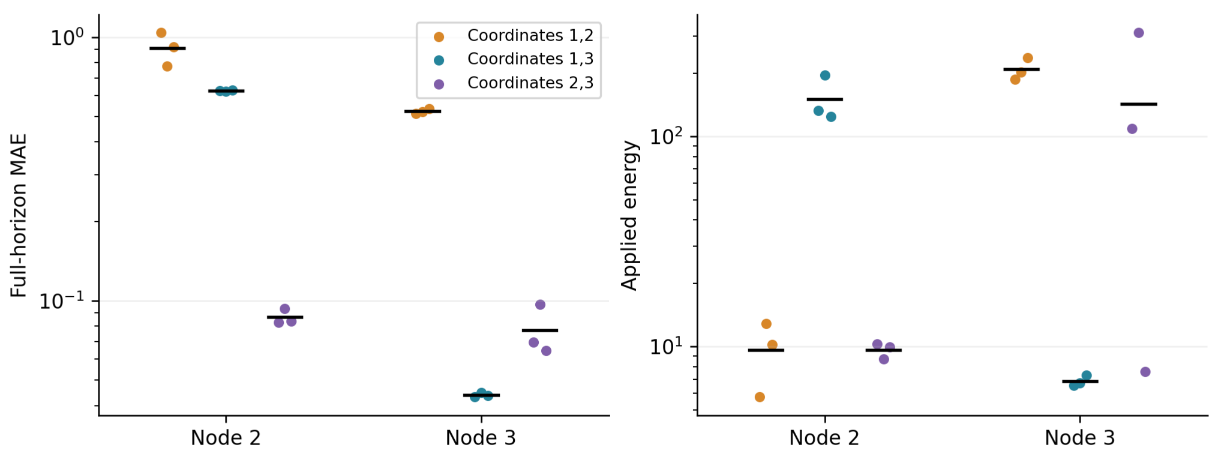


Figure 6: All paired four-dimensional subsets under single-node actuation. Each coordinate pair includes its drive states and error components. Points are individual training-run means over 20 common initial conditions; black marks average the three runs. Both error and energy are shown on logarithmic scales.

Table 5 and Fig. 6 confirm the actuator-dependent effect of coordinate composition. Node 2 performs best among these paired subsets with $\mathcal{O}_{23}$, whereas Node 3 favors $\mathcal{O}_{13}$ in both mean MAE and energy. The subsets omitting the actuated node's state–error pair, namely Node $2/\mathcal{O}_{13}$ and Node $3/\mathcal{O}_{12}$, give much larger residuals and energy than these favorable configurations. However, including that pair is not sufficient for low residuals: Node $2/\mathcal{O}_{12}$ remains poor. Thus, the identity of the accompanying coordinate also matters. These are finite-budget learning results, not a proof that local information is necessary or that a particular subset is intrinsically optimal. At Node 3, $\mathcal{O}_{23}$ also exhibits substantial energy variability across training runs; none is excluded.

4.2.2. *Accuracy-constrained configuration selection*

An offline selection procedure is evaluated using the original eight-candidate pool, fixed before the paired-subset completion. A candidate c specifies the actuated node and its observation subset. The pool comprises Nodes 2 and 3, each with local two-dimensional observations, the original four-dimensional subset ($\mathcal{O}_{12}$ at Node 2 and $\mathcal{O}_{13}$ at Node 3), $\mathcal{O}_{23}$, and full six-dimensional observations. Each candidate retains all three training runs; the procedure selects a configuration, not the best training run. The allowed actuator set may be $\{2,3\}$, $\{2\}$, or $\{3\}$.

For a trajectory, let z be the sample mean of $\|\mathbf{e}(t)\|_1$ over the final interval $16 \le t \le 20$. A trajectory passes when $z \le \epsilon$ and no state-protection boundary is reached. For training run r, its validation pass fraction is

$$\hat{p}_{c,r}(\epsilon) = \frac{1}{n_v} \sum_{\ell=1}^{n_v} \mathbf{1}\{z_{c,r,\ell} \le \epsilon,\ g_{c,r,\ell} = 0\}, \tag{44}$$

where g counts state-protection boundary hits. A candidate is feasible only if $\min_r \hat{p}_{c,r} \ge 0.90$. Among feasible candidates with allowed actuation, the rule first minimizes observation dimension and then validation mean applied energy E_u. Remaining ties are resolved by a fixed candidate ordering.

Table 6
Validation-selected configurations and independent-initial-condition tests. Test pass fractions are the minimum across three training runs, each evaluated on the same 40 test initial conditions. Error and energy average initial conditions and runs. A dash denotes no selection and hence no corresponding test; grouped tolerances have identical outcomes.

Allowed nodes	ε	Selected node / observations	Test pass	Test z	Test E_u
$\{2,3\}$	0.01	–	–	–	–
$\{2,3\}$	0.05, 0.10	$3\,/\,\mathcal{O}_{13}$	100%	0.0086	5.03
$\{2\}$	0.01, 0.05	–	–	–	–
$\{2\}$	0.10	$2\,/\,\mathcal{O}_{23}$	100%	0.0282	8.11
$\{3\}$	0.01	–	–	–	–
$\{3\}$	0.05, 0.10	$3\,/\,\mathcal{O}_{13}$	100%	0.0086	5.03

If none is feasible, no configuration is selected. This lexicographic preference treats feedback dimension as the primary resource, not as a measured hardware cost.

The rule uses 20 new common validation initial conditions and 40 separate common test initial conditions drawn from the original initial-state distribution. The illustrative tolerances $\varepsilon \in \{0.01, 0.05, 0.10\}$ were informed by the preceding experiments and fixed before this validation. They are neither application-derived requirements nor theoretical synchronization thresholds. Choices were recorded before evaluating the selected configurations on the test set, with no test-based reselection. Dynamics, nominal parameters, action bounds, and frozen policies were unchanged.

Table 6 translates the observation comparison into conditional design choices. With either node available, the original four-dimensional Node 3 configuration is selected at tolerances 0.05 and 0.10. When actuation is restricted to Node 2, the alternative four-dimensional subset is selected at 0.10. Both selected configurations pass every test trajectory at their respective selected tolerances. No candidate meets the validation criterion at 0.01. No state-protection boundary is reached in the 480 validation or 240 test trajectories.

The per-run criterion prevents aggregate averages from masking inconsistent training outcomes. At tolerance 0.05, the alternative Node 2 subset has validation pass fractions of 100%, 75%, and 100%; it is rejected despite an aggregate fraction above 90%. The local two-dimensional Node 3 policies likewise yield 100%, 0%, and 100% at both 0.05 and 0.10. Moreover, dimension-first selection is not energy minimization: at Node 2, the selected four-dimensional configuration has validation mean energy 11.76 versus 10.93 for six dimensions. The preference must therefore reflect the intended resource priority.

This is a candidate-limited, nominal-condition design procedure with held-out initial-state validation, not a globally optimal sensor–actuator design. The empirical pass fractions do not certify population reliability, new-training-run performance, or robustness to parameter changes. Shared initial conditions are not independent replications across policies. Finally, the mean-tail criterion does not require the error to remain below the tolerance at every instant, and no theorem equates these illustrative tolerances with application success.

Sensitivity to the selection preference. A post-hoc analysis reuses only the original eight-candidate validation data, varying the minimum per-run pass fraction over 80%, 90%, and 100%, and comparing dimension-first with energy-first ranking. Changing the pass-fraction requirement does not change any choice at the tested error tolerances. The ranking preference matters when only Node 2 is allowed at $\varepsilon = 0.10$: dimension-first selects $\mathcal{O}_{23}$, whereas energy-first selects full observations. All other choices are unchanged. This checks preference sensitivity, not a new held-out validation of altered rules. The subsequently completed paired subsets are not retroactively inserted into the original candidate set or its frozen choices.

Response-parameter mismatch. The original selected Node 2/$\mathcal{O}_{23}$ and Node 3/$\mathcal{O}_{13}$ policies are frozen and evaluated at their respective tolerances 0.10 and 0.05. For 40 new common initial conditions, response weights are set to $W_r = W \odot (1 + \eta U)$, where each entry of U is sampled uniformly from $[-1, 1]$ and $\eta \in \{0, 0.01, 0.03, 0.05\}$. The drive system is unchanged. Each initial condition uses the same mismatch direction across amplitudes and policies; an independent random stream preserves identical initial states. Zero weights remain zero. No retraining or configuration reselection is performed.

Table 7 shows that both configurations retain a minimum pass fraction of 97.5% at 1% relative mismatch, but fall below the original 90% requirement at 3% and 5%. No state-protection boundary is reached in these 960 trajectories. Hence nominal selection does not imply preservation of the requested accuracy under larger mismatch. The test identifies limits of the frozen policy–configuration combinations, rather than isolating a sensing mechanism or establishing a certified mismatch radius. It also does not assess retrained policies on different drive-network parameters.

4.2.3. *Actuator-count ablation*

To quantify the performance cost of restricting actuation to one node, nominal PPO synchronization is evaluated for every nonempty subset of the three nodes: $\{1\}$, $\{2\}$, $\{3\}$, $\{1,2\}$, $\{1,3\}$, $\{2,3\}$, and $\{1,2,3\}$. For a subset $\mathcal{S}$, the input matrix contains the corresponding columns of I_3, and the policy produces $|\mathcal{S}|$ actions. The six-dimensional observation, error-only reward, initial-state distribution, RK4

Table 7
Frozen selected configurations under response-weight mismatch. Pass fractions are the minimum across three training runs, each evaluated on the same 40 new initial conditions. Mean tail error z and energy E_u average runs and initial conditions. The two configurations use different, previously fixed tolerances; their pass fractions are not a matched-tolerance ranking.

Configuration	ε	η	Minimum pass	Mean z	Mean E_u
$2/\mathcal{O}_{23}$	0.10	0%	100.0%	0.0266	10.08
$2/\mathcal{O}_{23}$	0.10	1%	97.5%	0.0402	10.11
$2/\mathcal{O}_{23}$	0.10	3%	57.5%	0.0942	10.20
$2/\mathcal{O}_{23}$	0.10	5%	35.0%	0.1571	10.39
$3/\mathcal{O}_{13}$	0.05	0%	100.0%	0.0084	5.84
$3/\mathcal{O}_{13}$	0.05	1%	97.5%	0.0181	5.85
$3/\mathcal{O}_{13}$	0.05	3%	75.0%	0.0455	5.91
$3/\mathcal{O}_{13}$	0.05	5%	42.5%	0.0743	6.01

Table 8
Nominal PPO ablation over all nonempty actuator subsets. Each configuration uses three final policies and 20 common test initial conditions per policy. Values are mean ± SD across training-seed means; energy is summed over all active inputs.

Controlled nodes	Inputs	MAE	Steady L1 error	E_u
$\{1\}$	1	1.3937 ± 0.0029	4.2825 ± 0.1571	266.49 ± 13.41
$\{2\}$	1	0.0930 ± 0.0022	0.0426 ± 0.0131	9.09 ± 1.53
$\{3\}$	1	0.0431 ± 0.0012	0.0091 ± 0.0033	6.54 ± 0.34
$\{1,2\}$	2	0.1145 ± 0.0110	0.0679 ± 0.0212	20.61 ± 1.28
$\{1,3\}$	2	0.0232 ± 0.0007	0.0119 ± 0.0019	10.93 ± 0.12
$\{2,3\}$	2	0.0205 ± 0.0011	0.0141 ± 0.0040	8.15 ± 0.27
$\{1,2,3\}$	3	0.0118 ± 0.0007	0.0105 ± 0.0026	10.91 ± 0.17

integration, policy hidden layers, PPO hyperparameters, and nominal training budget of 4×10^5 transitions are unchanged. Each configuration includes three independent training runs and is evaluated on 20 common test initial conditions. All seven configurations and all three final checkpoints per configuration are reported, including configurations with poor synchronization.

The same per-actuator bound $|u_j| \leq 4$ is imposed in every configuration, while E_u sums energy across all active channels. Thus, additional actuators provide both additional control directions and a larger admissible total input magnitude ($\|\mathbf{u}\|_2 \leq 4\sqrt{|\mathcal{S}|}$). This is an actuator-availability comparison, not a comparison at fixed aggregate control authority. Only nominal synchronization is tested in this ablation. Actuator count measures the structural actuation requirement; hardware cost and implementation overhead are not measured.

Table 8 and Fig. 7 quantify the tradeoff. Single-node control at Node 3 gives MAE 0.0431 and energy 6.54, whereas $\{2,3\}$ gives 0.0205 and 8.15 and $\{1,2,3\}$ gives 0.0118 and 10.91. Thus, one actuator retains low-residual synchronization with lower measured energy, but incurs a higher full-horizon error than these multi-input configurations. The steady L1 error does not decrease monotonically with actuator count: it is 0.0091 for $\{3\}$, compared with 0.0141 for $\{2,3\}$ and 0.0105 for $\{1,2,3\}$. The full-horizon improvement should therefore not be described as uniformly better steady-state precision.

The $\{1,2\}$ configuration has higher MAE and energy than $\{2\}$ under the fixed training budget, while configurations containing Node 3 perform better in full-horizon MAE. This emphasizes the effect of actuator placement and finite-budget policy optimization. Since adding an actuator enlarges the admissible control set, the poorer learned result for $\{1,2\}$ is not evidence that the extra actuator intrinsically reduces controllability. These results support single-input control as a resource-constrained design choice, with an explicit performance cost, rather than as the best-performing architecture for every objective.

4.3. Robustness under Parameter Mismatch and External Perturbations

4.3.1. Parameter mismatch

After assessing actuator placement and observation availability, the frozen controllers are evaluated under parameter mismatch [30]. To model uncertainty in the response system, a mismatch is introduced into the weight matrix. Let W be the weight matrix of the drive system. The mismatched weight matrix of the response system is constructed as

$$W_r = W + \eta(W \odot \Delta), \tag{45}$$

where η denotes the mismatch intensity coefficient, $\odot$ denotes the Hadamard product, and $\Delta = [\delta_{ij}]$ is a random perturbation matrix with

$$\delta_{ij} \sim U(-0.03, 0.03). \tag{46}$$

This formulation corresponds to a relative multiplicative mismatch. That is, each nonzero connection weight in the

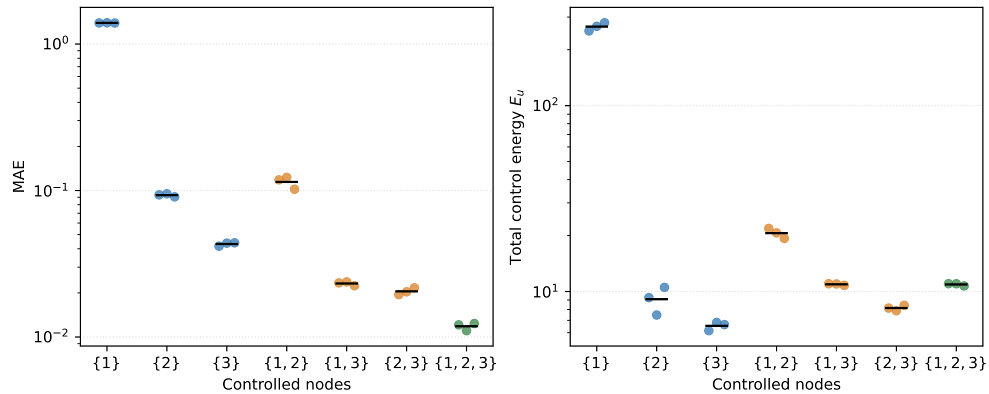


Figure 7: Nominal synchronization error and total control energy for all actuator subsets. Each colored point is a training-seed mean over 20 common test initial conditions; black horizontal marks denote the mean across three seeds. Blue, orange, and green indicate one, two, and three actuators, respectively.

response system is perturbed around its nominal value, while zero entries remain unchanged. This modeling approach preserves the original network topology and is also consistent with practical parameter deviations caused by manufacturing errors, calibration bias, and component tolerances.

It should be noted that η is a normalized scaling coefficient rather than the direct percentage of mismatch. When $\eta = 1.0$, the relative perturbation range is approximately $\pm 3\%$ of the original weight; when $\eta = 2.0$, the range is approximately $\pm 6\%$. In this work, the following mismatch levels are considered:

$$\eta \in \{0.0, 0.5, 1.0, 1.5, 2.0\}. \tag{47}$$

Here, $\eta = 0.0$ corresponds to the nominal case without mismatch, while larger values represent stronger parameter uncertainty.

All controllers are evaluated without adaptation under the perturbed response matrices. PPO is trained without domain randomization; LQR is designed from the nominal linearization, and SMC uses nominal compensation with parameters selected on separate validation scenarios as described above.

Fig. 8 reports MAE and control energy for the reference controller settings. Table 9 gives endpoint results and the impulse metric discussed below. At Node 2, the unchanged $k = 8$ feedback has large errors; at $\eta = 2$, PPO attains MAE 0.1703, compared with 0.2330 for the $R = 1$ LQR and 0.2223 for SMC, at higher energy than both. Node 3 generally shows smaller absolute error differences. These comparisons depend on the specified baseline parameters; Section 4.3.4 shows that a different validated LQR weighting reverses several error comparisons.

4.3.2. Impulse disturbance

In addition to parameter uncertainty, sudden impulse disturbance is also considered to evaluate the transient recovery capability of the controllers [31]. In the testing implementation, an impulse is added to all three drive-state components immediately after integration at step 200 ($t = 10$); the response state is not directly shifted. This tests recovery following an abrupt change of the synchronization reference. The impulse vector is set as

$$\mathbf{p} = [5, 5, 5]^{\mathrm{T}}. \tag{48}$$

The same drive-state jump is applied for both actuator locations and all controllers. It tests recovery after a shared reference change; the resulting response can still depend on the chosen jump direction.

Because the immediate peak M_p is dominated by the prescribed jump, E_{post} is used as the main recovery metric. Among the reference settings in Fig. 9 and Table 9, PPO has the lowest Node 2 mean (2.0800), versus 2.5264 for LQR with $R = 1$, 2.7691 for SMC, and 8.6188 for fixed feedback. PPO uses more energy than the first two baselines. At Node 3, fixed feedback and PPO have similar means (1.2518 and 1.2969), while the reference LQR uses less energy. These statements are restricted to the displayed settings; the selected LQR weighting and SAC comparison below alter the Node 2 error ordering.

4.3.3. Gaussian-noise perturbation

To further evaluate the robustness of the learned controller under persistent random perturbations, zero-mean Gaussian noise is added to the response system during testing, while the training stage remains noise-free. This setting tests the frozen policy under stochastic state perturbations absent from training. Specifically, after each control step and numerical integration, additive Gaussian noise is imposed on the response state:

$$\mathbf{y}_{k+1}^{\mathrm{noisy}} = \mathbf{y}_{k+1} + \boldsymbol{\xi}_k, \tag{49}$$

where

$$\boldsymbol{\xi}_k \sim \mathcal{N}(\mathbf{0}, \sigma^2 I_3). \tag{50}$$

Here, σ denotes the noise intensity and I_3 is the 3×3 identity matrix. The tested noise levels are

$$\sigma \in \{0, 0.01, 0.02, 0.05\}. \tag{51}$$

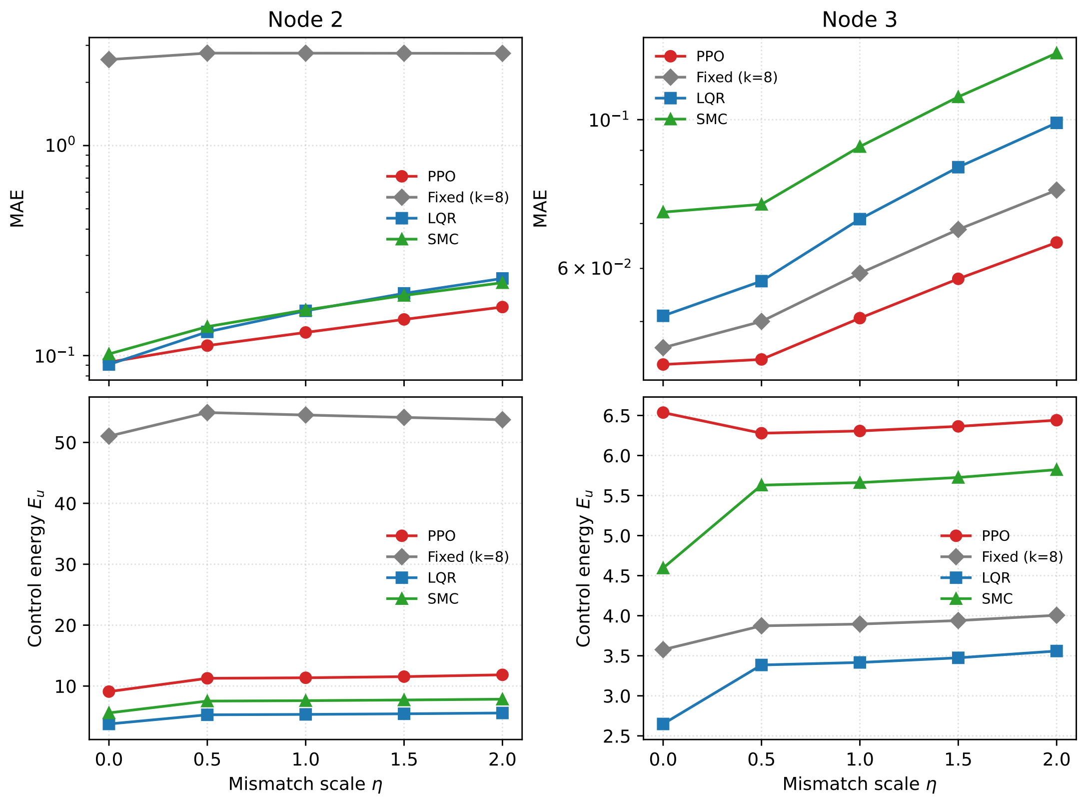


Figure 8: MAE and control energy of PPO, fixed-gain feedback ($k = 8$), LQR, and SMC under parameter mismatch. Curves show the mean over the common test initial conditions; PPO additionally aggregates three training seeds.

Table 9
Representative mismatch and impulse-disturbance results.

Node	Method	Nominal		$\eta = 2$	Impulse	
		MAE	E_u	MAE	E_{post}	E_u
2	PPO	0.0930	9.09	0.1703	2.0800	17.51
	Fixed ($k = 8$)	2.5673	51.03	2.7481	8.6188	74.02
	LQR	0.0907	3.78	0.2330	2.5264	9.95
	SMC	0.1018	5.59	0.2223	2.7691	14.13
3	PPO	0.0431	6.54	0.0656	1.2969	18.09
	Fixed ($k = 8$)	0.0457	3.58	0.0785	1.2518	16.21
	LQR	0.0510	2.65	0.0989	1.6883	7.49
	SMC	0.0728	4.60	0.1257	1.7352	11.37

The noise experiment is conducted for both pinning nodes. Fig. 10 shows increasing MAE with σ for each reference controller. At $\sigma = 0.05$, PPO has the smallest mean MAE among the four settings in Table 10, which uses LQR with $R = 1$. For Node 2 its MAE is 0.2669, versus 0.3742 for that LQR and 0.3295 for SMC, but with substantially greater energy than LQR. This ranking does not include the selected $R = 0.01$ LQR or SAC, which achieve lower mean MAE in the subsequent comparisons. Noise is added to the evolving response state, so this is a process-noise experiment, not a measurement-only test or an arbitrary-noise robustness guarantee.

4.3.4. Sensitivity to LQR control weighting

The preceding LQR curves use the reference design $Q = I_3$, $R = 1$. To check whether the error comparisons depend on that choice, a separate sensitivity test fixes $Q = I_3$ and considers $R \in \{0.01, 0.1, 1, 10, 100\}$. For each node, R is selected using ten validation initial conditions and the same nominal, $\eta = 1$, and impulse normalized-MAE score used for SMC. All five candidate scores are retained in the reproducibility records. Both nodes select $R = 0.01$, which is then frozen before evaluating 20 held-out test initial conditions. This error-oriented validation does not penalize measured energy and has access to non-ideal validation

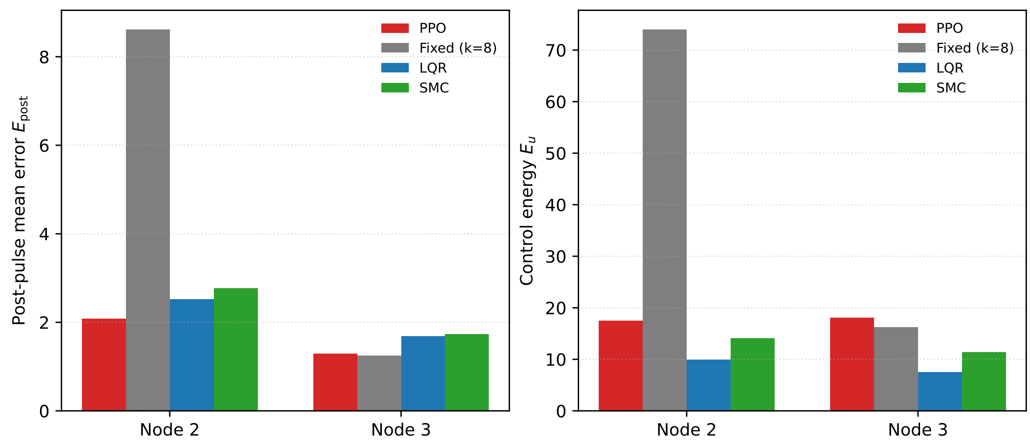


Figure 9: Post-pulse mean error and control energy for the reference settings under the same global impulse. LQR uses $R = 1$.

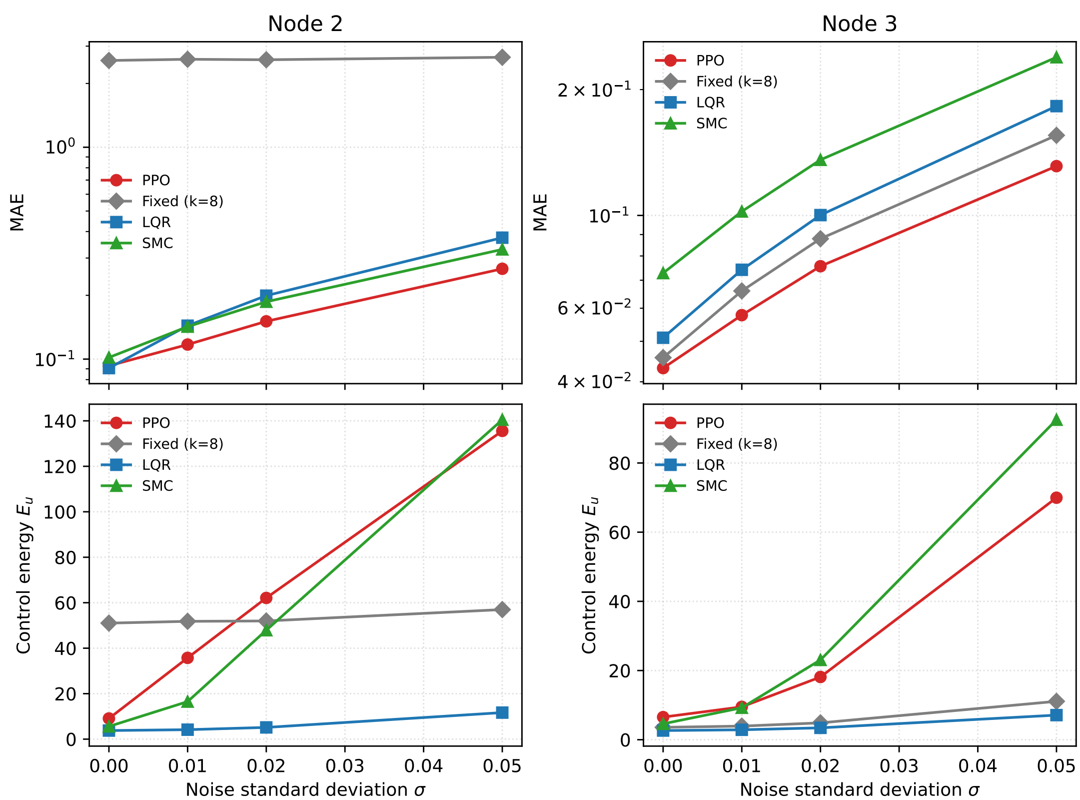


Figure 10: MAE and control energy of the four reference controllers under Gaussian-noise perturbation. LQR uses $R = 1$.

scenarios, unlike nominal-only PPO training. The selected value is at the lower edge of this limited grid; it is not claimed to be globally optimal. The historical $R = 1$ results and fixed gain $k = 8$ are unchanged.

Table 11 shows that the original LQR weighting materially affects the comparison. At Node 2, $R = 0.01$ lowers MAE below PPO in all four tested settings. Under mismatch, its MAE and energy are 0.1486 and 14.60, compared with PPO's 0.1703 and 11.85. After the impulse, its mean post-pulse error is 1.5536 versus PPO's 2.0800, with energy 31.23 versus 17.51. Under noise, it improves both MAE (0.2427 versus 0.2669) and energy (90.27 versus 135.57). Thus, PPO's error advantages over the reference $R = 1$ design must not be generalized to tuned LQR.

At Node 3, the selected LQR has lower mean error and energy than PPO in the nominal, mismatch, and noise tests. PPO has a slightly lower mean post-impulse error (1.2969 versus 1.3169), but higher energy (18.09 versus 13.71); this small difference is not interpreted as statistical superiority. The result reinforces the error–effort interpretation and the importance of controller weighting, rather than establishing an inherent robustness advantage of learning-based control.

Table 10
Performance at the highest tested noise intensity, $\sigma = 0.05$.

Node	Method	MAE	Steady L1 error	E_u
2	PPO	0.2669	0.5383	135.57
	Fixed ($k = 8$)	2.6584	8.7280	56.99
	LQR	0.3742	0.8979	11.67
	SMC	0.3295	0.7298	140.49
3	PPO	0.1312	0.3183	69.96
	Fixed ($k = 8$)	0.1555	0.4000	11.09
	LQR	0.1826	0.4419	7.11
	SMC	0.2389	0.6029	92.51

Table 11
LQR weight sensitivity on held-out tests. Each entry is a mean over 20 initial conditions; PPO additionally averages three training seeds. The selected $R = 0.01$ is frozen using separate validation seeds. MAE covers the full horizon, including the impulse scenario.

		PPO		LQR, $R = 1$		LQR, $R = 0.01$	
Node	Scenario	MAE	E_u	MAE	E_u	MAE	E_u
2	Nominal	0.0930	9.09	0.0907	3.78	0.0772	10.58
2	$\eta = 2$	0.1703	11.85	0.2330	5.58	0.1486	14.60
2	Impulse	0.4339	17.51	0.5128	9.95	0.3368	31.23
2	$\sigma = 0.05$	0.2669	135.57	0.3742	11.67	0.2427	90.27
3	Nominal	0.0431	6.54	0.0510	2.65	0.0392	4.99
3	$\eta = 2$	0.0656	6.44	0.0989	3.56	0.0647	5.29
3	Impulse	0.2581	18.09	0.3330	7.49	0.2592	13.71
3	$\sigma = 0.05$	0.1312	69.96	0.1826	7.11	0.1264	23.27

4.4. Comparison with Soft Actor-Critic

To assess the dependence of the results on the learning algorithm, soft actor-critic (SAC) [40] is evaluated on the same single-input Node 2 and Node 3 tasks. Both algorithms use the six-dimensional observation $(\mathbf{x}, \mathbf{e})/5$, reward $-0.1\|\mathbf{e}\|_1$, actuator bound $|u| \leq 4$, nominal training dynamics, and final checkpoints at a nominal budget of 4×10^5 environment transitions. Three independent training runs and 20 common test initial conditions per seed are used. All checkpoints are retained, and evaluation uses deterministic actions without online learning. The compared test settings are nominal dynamics, mismatch $\eta = 2$, the drive-state impulse at $t = 10$, and response process noise with $\sigma = 0.05$.

Both implementations use two hidden layers of width 64, a learning rate of 3×10^{-4}, batch size 128, and discount factor 0.99. PPO uses Tanh hidden activations, four environments with 512 steps per rollout, ten optimization epochs, clip range 0.2, GAE parameter 0.95, and entropy coefficient 0.001. SAC uses ReLU hidden activations, one environment, a replay buffer of 10^5 transitions, 5,000 initial exploration steps, one gradient update per subsequent transition, target-update coefficient $\tau = 0.005$, and automatic entropy adjustment with target entropy -1. These settings are fixed across nodes and seeds. Equal environment-transition budgets do not imply equal gradient-update counts or training times, and this comparison does not exhaust algorithm-specific hyperparameter tuning.

Table 12 and Fig. 11 show a scenario-dependent comparison. For Node 2, PPO has lower nominal MAE (0.0930 versus 0.1173), but SAC has lower MAE under mismatch $\eta = 2$ (0.1513 versus 0.1703) and also lower energy (9.31 versus 11.85). Following the impulse, SAC has lower mean E_{post} (1.9226 versus 2.0800), but greater variation across training-seed means (SD 0.5913 versus 0.1451) and higher mean energy (25.20 versus 17.51). At $\sigma = 0.05$, the mean MAEs are close (0.2632 for SAC and 0.2669 for PPO), while SAC uses less energy.

For Node 3, SAC has slightly lower mean MAE in the nominal, mismatch, and noise tests and lower mean energy in all four settings. PPO achieves lower post-impulse error (1.2969 versus 1.3987), at higher energy. Thus, the traditional-baseline advantages reported above do not imply that PPO outperforms another learning algorithm in every scenario. PPO is a viable implementation of the constrained pinning framework, while algorithm choice depends on the operating condition and the relative importance of error, variability, and control effort. With only three training seeds, the reported means and SDs are descriptive evidence rather than a general statistical superiority claim.

4.5. Design Implications and Scope

The central empirical finding is that observation composition matters alongside actuator placement. The matched-dimension comparison recovers low residuals at Node 2 without restoring all six inputs, whereas the same alternative subset increases residuals and effort at Node 3. Thus, the original Node 2 degradation cannot be explained by reduced input dimension alone. These within-node comparisons support configuration-specific evaluation of sensing and actuation, not an intrinsic observability mechanism or an optimal sensor choice; finite-budget policy optimization remains part of the measured outcome. The selection study in Section 4.2.2 gives this finding an operational interpretation: the preferred candidate depends on the allowed actuator location and the requested error tolerance, while stringent requirements may leave no feasible candidate. Its dimension-first ranking is an explicit design preference, not evidence of simultaneous optimality in precision and energy. Post-hoc ranking checks and frozen-policy mismatch tests expose the preference dependence and the limited transfer of nominal accuracy requirements.

Accuracy and applied effort can be competing requirements in control design [41]. Here they are evaluated jointly, but the error-only PPO reward is not a multi-objective learning formulation. The actuator ablation shows lower full-horizon error for favorable multi-input policies than for single-input Node 3, at greater measured total effort; steady-state error need not improve. Since per-channel limits are fixed, the multi-input cases also have greater aggregate control authority. Traditional and SAC comparisons establish the range of the PPO results under the tested designs, rather

Table 12
PPO and SAC at the same nominal budget of 4×10^5 environment transitions. Values are mean $\pm$ sample SD across three training-seed means, each obtained from 20 common test initial conditions. E_{post} is reported only for the impulse scenario.

Node	Scenario	Method	MAE	E_u	E_{post}
2	Nominal	PPO	0.0930 ± 0.0022	9.09 ± 1.53	–
2	Nominal	SAC	0.1173 ± 0.0046	6.27 ± 0.45	–
2	$\eta = 2$	PPO	0.1703 ± 0.0085	11.85 ± 1.60	–
2	$\eta = 2$	SAC	0.1513 ± 0.0037	9.31 ± 0.43	–
2	Impulse	PPO	0.4339 ± 0.0246	17.51 ± 1.19	2.0800 ± 0.1451
2	Impulse	SAC	0.4206 ± 0.0979	25.20 ± 9.79	1.9226 ± 0.5913
2	$\sigma = 0.05$	PPO	0.2669 ± 0.0049	135.57 ± 48.71	–
2	$\sigma = 0.05$	SAC	0.2632 ± 0.0052	86.13 ± 8.86	–
3	Nominal	PPO	0.0431 ± 0.0012	6.54 ± 0.34	–
3	Nominal	SAC	0.0416 ± 0.0010	4.60 ± 0.11	–
3	$\eta = 2$	PPO	0.0656 ± 0.0025	6.44 ± 0.30	–
3	$\eta = 2$	SAC	0.0644 ± 0.0009	5.07 ± 0.16	–
3	Impulse	PPO	0.2581 ± 0.0003	18.09 ± 3.50	1.2969 ± 0.0070
3	Impulse	SAC	0.2735 ± 0.0106	12.62 ± 0.53	1.3987 ± 0.0664
3	$\sigma = 0.05$	PPO	0.1312 ± 0.0066	69.96 ± 17.55	–
3	$\sigma = 0.05$	SAC	0.1273 ± 0.0013	50.40 ± 14.21	–

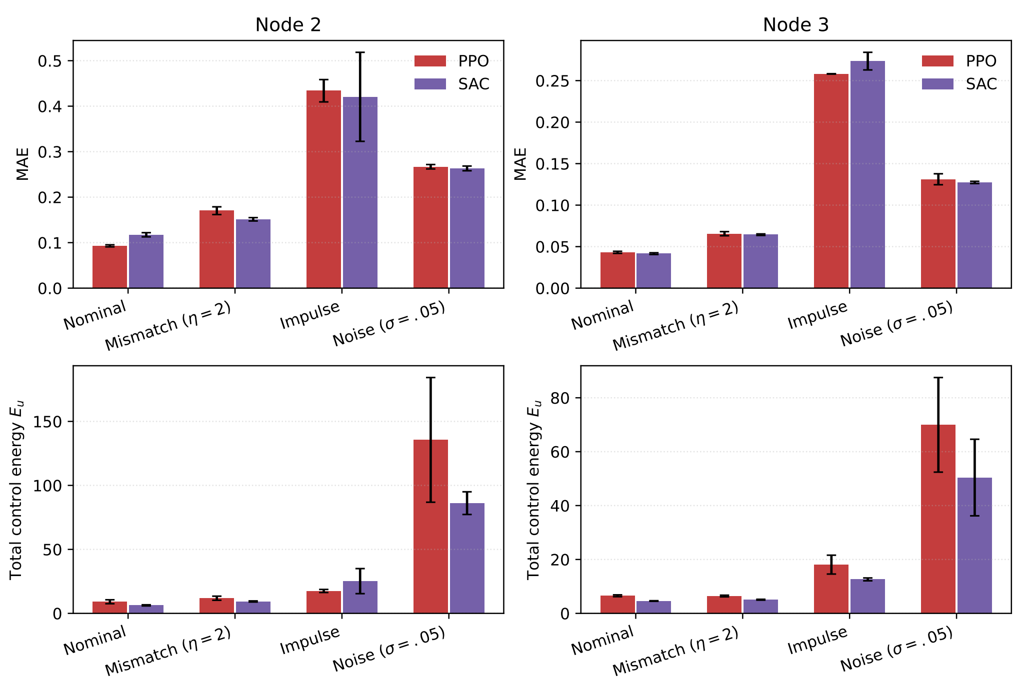


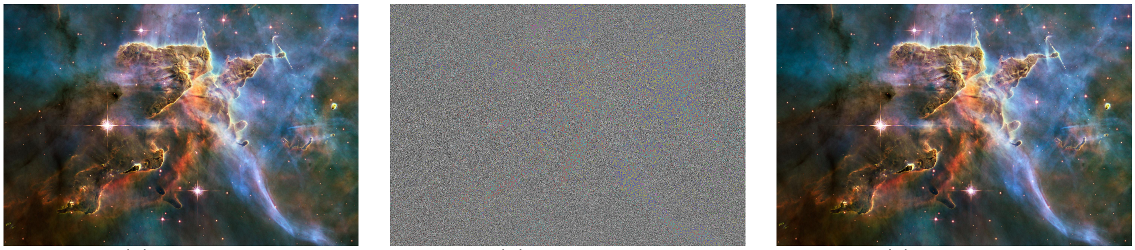
Figure 11: PPO–SAC comparison at matched environment-transition budgets. Bars show the mean of three training-seed means and error bars show their sample standard deviation. Each seed is evaluated on 20 common initial conditions. The upper panels use full-horizon MAE for every scenario; the separate post-impulse metric is given in Table 12.

than a controller-independent ranking of the nodes or an overall algorithm winner.

The scope is the stated three-state network, initial-state distribution, and separately applied perturbations. The actuator ablation fixes per-channel limits, allowing total available input magnitude to increase with actuator count. The observation study concerns policy inputs at deployment; its training reward uses all error components. Three training seeds describe variability within the reported protocol. Transfer to other networks, combined disturbances, certified synchronization bounds, and measured hardware costs remain subjects for further work.

5. Image Encryption Scheme and Statistical Security Analysis

The legacy implementation and its existing image statistics are described first. Section 5.2 then introduces a separate,



(a) Original (b) Encrypted (c) Decrypted

Figure 12: The original, encrypted, and decrypted images.

conditionally justified reconstruction experiment to examine the missing link between continuous synchronization error and discrete rule agreement; its results must not be attributed to the legacy preprocessing.

5.1. Legacy Image-Transform Implementation

Astronomical systems generate large amounts of high-value image data, including celestial structures, planetary surfaces, and remote-sensing information. In distributed observation scenarios, such as Very Long Baseline Interferometry [42], images may be transmitted between observation sites. This creates exposure to eavesdropping, tampering, and unauthorized access, making secure image protection important for astronomical and space-related applications.

Astronomical images usually contain rich textures, complex intensity distributions, and strong spatial correlations among adjacent pixels, making them suitable test objects for image encryption algorithms. An effective encryption scheme should conceal the visual content of the original image and reduce its statistical redundancy. Since chaotic systems are sensitive to initial conditions and can generate pseudo-random trajectories, they have been widely used for image encryption. In a drive–response synchronization framework, synchronized chaotic trajectories can provide consistent key streams for encryption and decryption.

Motivated by this idea, this section applies synchronized trajectories from the PPO-controlled continuous-time Hopfield chaotic neural network to image encryption. Figure 13 shows the workflow. The scheme combines synchronized Hopfield chaotic sequences, Logistic-map-based masking, and dynamic DNA coding. The plain image is encrypted and then decrypted to verify reversibility. RGB histograms and adjacent-pixel correlations are further analyzed to evaluate the statistical properties of the encrypted image. The specific steps are as follows:

1. *Channel separation and spatial blocks.* Let $P \in \{0,\dots,255\}^{H\times W\times 3}$ be a three-channel image. Each channel is sliced into non-overlapping blocks of edge length $p = 8$, visited from left to right and then from top to bottom. Boundary blocks may be smaller. Thus, the number of blocks is

$$N_b = \lceil H/p \rceil \lceil W/p \rceil. \tag{52}$$

Write $B_{c,b} = \mathcal{F}_p(P_c)_b$, where $\mathcal{F}_p$ denotes this ordered slicing operation, not a permutation of pixels. Its inverse places each block back at its original location.

2. *Image-dependent Logistic mask.* The implemented image feature is the sum of all channel values,

$$S(P) = \sum_{c=1}^{3}\sum_{i=1}^{H}\sum_{j=1}^{W} P(i,j,c). \tag{53}$$

Let d be the decimal string representing $S(P)$. The password string is concatenated directly with d, encoded in UTF-8, and hashed using SHA-256. Interpreting the digest as a nonnegative integer h, the implementation initializes a scalar floating-point value $z_0 = h/2^{256}$. The endpoint values zero and one, if obtained after floating-point conversion, are replaced by two const float, respectively. The sequence is then generated by

$$z_{t+1} = 3.9999 z_t(1-z_t), \qquad t = 0,\dots,HW-2. \tag{54}$$

With rnd denoting rounding to nearest with ties to even, as in NumPy, the mask is

$$Q = \mathrm{reshape}_{H\times W}\left((\mathrm{rnd}(10^4 z_t) \bmod 256)_{t=0}^{HW-1} \right). \tag{55}$$

The reshape uses row-major order. This is a floating-point mask generator; the hash length is not a claim about its effective key space. The receiver needs the same password and d to reconstruct Q.

3. *Channelwise masking.* Each channel uses the same mask, partitioned by the same operation:

$$D_{c,b} = B_{c,b} \oplus \mathcal{F}_p(Q)_b. \tag{56}$$

Here $\oplus$ is bytewise XOR. The implementation does not contain a circular or sequential cross-channel diffusion stage.

4. *Rule-stream construction.* After the trajectory transient, one sample is collected for each block. From the three drive states, four scalar streams are formed as

$$h_1 = x_1, \quad h_2 = x_2, \quad h_3 = x_3, \quad h_4 = x_1x_2 + x_3. \tag{57}$$

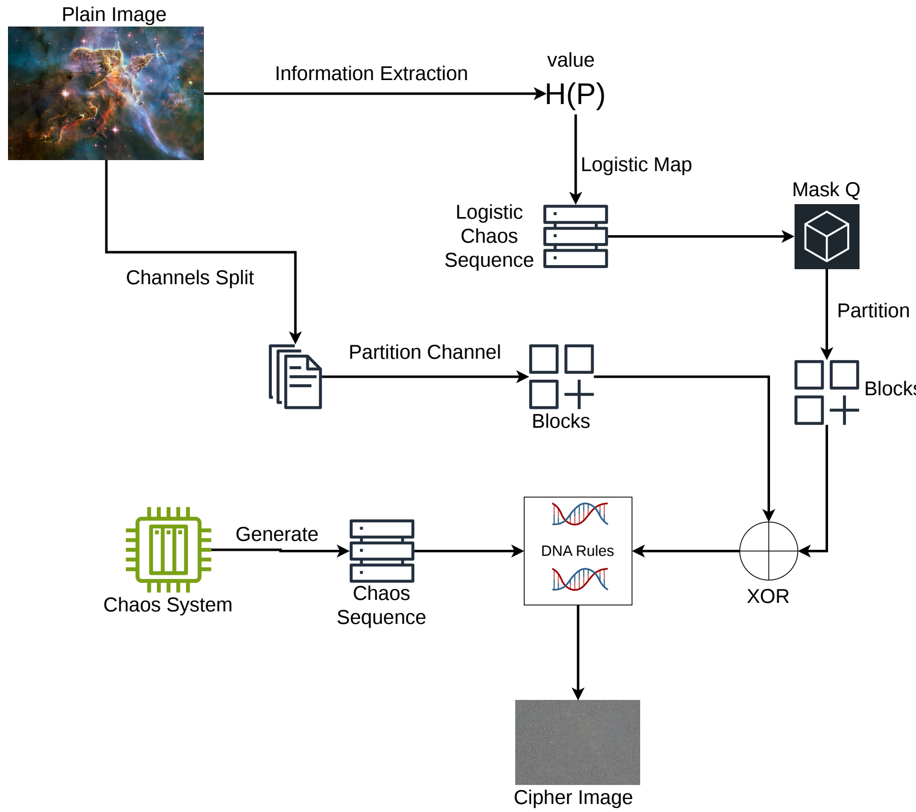


Figure 13: Overview of the image-encryption workflow. The image-derived value denoted by $H(P)$ in the schematic is combined with the password and mapped through SHA-256 to the Logistic initial value, as specified in the text. Chaotic sequences determine the discrete DNA rules after preprocessing. This overview omits the round-level operations and receiver-side reconstruction; channelwise XOR masking is not cross-channel diffusion.

The fourth stream is derived algebraically, not an additional dynamical state. Response streams use the corresponding y states. For each stream, the implemented preprocessing is

$$q^{(0)} = (\mathrm{rnd}(h) \bmod 8) + 1, \tag{58}$$

$$q^{(\ell+1)} = \left\lfloor (\mathcal{K}[q^{(\ell)}] \bmod 8) + 1 \right\rfloor. \tag{59}$$

The second update is repeated for $\ell = 0, \ldots, 14$. Here $\mathcal{K}$ is the scalar Kalman filter implemented as a forward pass, with measurement variance 10^{-2} and process variance 10^{-5}. Each pass is restarted with the first input element as its estimate and measurement variance as its initial error variance. The floor represents the implemented conversion of positive values to unsigned integers. The resulting indices are $s_{i,b} = q_{i,b}^{(15)} \in \{1, \ldots, 8\}$. This legacy preprocessing is not a dead-zone quantizer and does not provide a bound guaranteeing identical indices from small continuous-state errors.

5. *Two-round DNA substitution and reconstruction.* Let R_i be the bijection in Table 13, applied to each consecutive two-bit pair of a byte, starting from the most significant pair. The same block index is used for all pixels of that block and all three channels:

$$D'_{c,b} = R^{-1}_{s_{2,b}}(R_{s_{1,b}}(D_{c,b})), \tag{60}$$

$$D''_{c,b} = R^{-1}_{s_{4,b}}(R_{s_{3,b}}(D'_{c,b})), \tag{61}$$

$$C_c = \mathcal{F}_p^{-1}((D''_{c,b})_{b=1}^{N_b}). \tag{62}$$

These operations are pointwise substitutions, not spatial scrambling or propagation between pixels.

Table 13
DNA coding rules used in the implementation.

Rule	00	01	10	11
1	A	G	C	T
2	A	C	G	T
3	T	G	C	A
4	T	C	G	A
5	C	A	T	G
6	G	A	T	C
7	C	T	A	G
8	G	T	A	C

For decryption, the response indices are applied in reverse order: encode with the fourth rule, decode with the third, encode with the second, and decode with the first.

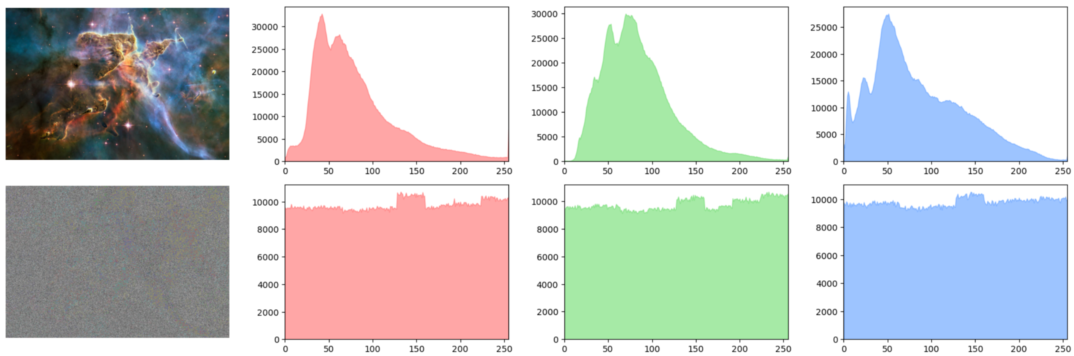


Figure 14: RGB histogram comparison between the plain image and the cipher image.

The result is XORed with the same mask and the blocks are reassembled. Identical drive and response indices, together with identical Q, are sufficient for exact recovery because each DNA map is bijective and XOR is self-inverse. This conditional reversibility does not establish that small synchronization errors yield matching indices, nor that the image transform is cryptographically secure.

The current in-process interface returns both d and the Logistic initial value, but decryption uses only d and recomputes the initial value from the password. This return value is not a specified public communication protocol; in particular, disclosure of the initial value would disclose the mask generator's seed. Reliability of independently generated rule streams and secure exchange of auxiliary information require separate evaluation.

To illustrate the effectiveness of the proposed encryption method, we apply it to a color astronomical image. As shown in Fig. 12(a) and Fig. 12(b), the encrypted image exhibits a noise-like appearance and has no obvious visual similarity to the original image, indicating that the visual information is effectively concealed. Fig. 12(c) shows that the decrypted image is visually consistent with the original image, which illustrates a successful recovery example. Exact recovery is assessed by pixelwise equality rather than visual similarity or the implementation's return flag.

After visual inspection, histogram and adjacent-pixel correlation analyses are used to further evaluate the statistical properties of the cipher image. For each color channel $c \in \{R, G, B\}$, the histogram is defined as

$$H_c(k) = \#\{(i,j) \mid I_c(i,j) = k\}, \qquad k = 0, 1, \dots, 255. \tag{63}$$

As shown in Fig. 14, the histograms of the plain image are highly non-uniform, whereas those of the cipher image are much flatter over the gray-level range. This indicates that the encryption process weakens the original pixel-intensity distribution and reduces histogram-based statistical leakage.

The adjacent-pixel correlation coefficient is computed as

$$r_{xy} = \frac{\mathrm{cov}(X,Y)}{\sqrt{D(X)}\sqrt{D(Y)}}, \tag{64}$$

where

$$\mathrm{cov}(X,Y) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(X))(y_i - E(Y)), \tag{65}$$

and

$$D(X) = \frac{1}{N}\sum_{i=1}^{N}(x_i - E(X))^2. \tag{66}$$

Here, X and Y denote two adjacent-pixel sequences sampled in the horizontal, vertical, or diagonal direction. As reported in Table 14, the original images have correlation coefficients close to 1 in all three directions, confirming the strong spatial redundancy of natural images. After encryption, the coefficients produced by both the node2 and node3 models decrease to values close to 0. The scatter plots in Fig. 15 also show that the encrypted pixels are broadly dispersed instead of being concentrated along the diagonal line. These results indicate that the proposed encryption process effectively suppresses adjacent-pixel correlation.

It should be noted that the node1 model is not included because its synchronization performance is insufficient, causing the generated master–slave key streams to fail the sequence consistency check required for correct encryption and decryption. Both node2 and node3 are evaluated under the six-dimensional full-observation setting. Compared with the encrypted images reported in Refs. [43, 44], the proposed method achieves comparable correlation suppression, demonstrating its effectiveness in reducing local statistical redundancy.

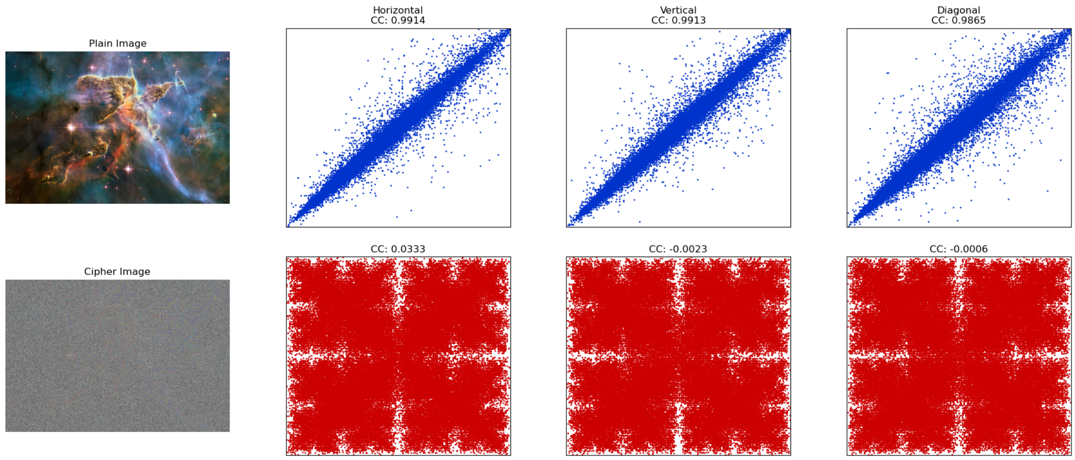


Figure 15: Adjacent-pixel correlation analysis of the plain image and the cipher image in horizontal, vertical, and diagonal directions.

Table 14
Correlation coefficients of the original and encrypted images

Image	Horizontal	Vertical	Diagonal
Original image of node2 model	0.9914	0.9913	0.9865
Encrypted image of node2 model	0.0333	-0.0023	-0.0006
Original image of node3 model	0.9921	0.9917	0.9852
Encrypted image of node3 model	0.0381	-0.0018	-0.0012
Encrypted image of Ref. [43]	-0.0015	0.0023	0.0021
Encrypted image of Ref. [44]	-0.0123	-0.0068	-0.0368

5.2. From Synchronization Error to Discrete Recovery

The control analysis and image application address different requirements. Section 3.3 establishes state boundedness and supplies sampled evidence of local perturbation attenuation, but neither property alone guarantees identical discrete rule streams. For example, rounding 0.49 and 0.51 produces different integers despite their small separation. To make this connection explicit, the following conditional construction is evaluated separately from the preceding legacy image transform and its statistical figures.

Write $h(\mathbf{x}) = (x_1, x_2, x_3, x_1x_2 + x_3)^{\mathsf{T}}$. At each block sample b, the sender uses a fixed width $\Delta > 0$ and forms, componentwise,

$$\begin{aligned} k_{i,b} &= \left\lfloor h_i(\mathbf{x}_b)/\Delta + \tfrac{1}{2} \right\rfloor, \\ r_{i,b} &= h_i(\mathbf{x}_b)/\Delta - k_{i,b}, \\ s_{i,b} &= (k_{i,b} \bmod 8) + 1. \end{aligned} \tag{67}$$

The receiver receives the aligned centering offsets $r_{i,b}$ as auxiliary information, not the sender's integers, and computes

$$\widehat{k}_{i,b} = \left\lfloor h_i(\mathbf{y}_b)/\Delta - r_{i,b} + \tfrac{1}{2} \right\rfloor, \quad \widehat{s}_{i,b} = (\widehat{k}_{i,b} \bmod 8) + 1. \tag{68}$$

Proposition 5.1 (Conditional discrete agreement and recovery)**.** *Assume aligned samples, exact arithmetic, unaltered auxiliary information, and the same byte mask at both ends. If*

$$D = \max_b \|h(\mathbf{y}_b) - h(\mathbf{x}_b)\|_\infty < \Delta/2, \tag{69}$$

then all reconstructed rule indices equal the sender's indices. Any blockwise bijective substitution indexed by these rules is exactly reversible. In particular, if $|x_{j,b}|, |y_{j,b}| \le M$ *for* $j = 1, 2$ *and* $\max_b \|\mathbf{y}_b - \mathbf{x}_b\|_\infty \le \varepsilon$*, a sufficient state-error condition is* $(2M + 1)\varepsilon < \Delta/2$.

Proof. Substituting (67) into (68) gives $\widehat{k}_{i,b} = k_{i,b} + \lfloor (h_i(\mathbf{y}_b) - h_i(\mathbf{x}_b))/\Delta + 1/2 \rfloor$. Under (69), the last floor is zero. Thus $\widehat{s}_{i,b} = s_{i,b}$; inverse substitution followed by XOR with the identical mask recovers each original byte. For the state-error condition, let $\mathbf{e}_b = \mathbf{y}_b - \mathbf{x}_b$. The fourth-stream difference is $y_{2,b}e_{1,b} + x_{1,b}e_{2,b} + e_{3,b}$, whose magnitude is at most $(2M + 1)\varepsilon$; the other three differences are bounded by ε. □

State boundedness supplies finite amplitude bounds, but the required uniform ε is not proved for PPO in Section 3.3. Accordingly, the application condition is checked on the collected trajectories, not inferred from a negative Lyapunov estimate or a mean error. In finite precision, a positive margin must also cover stream-evaluation, offset-storage, and rounding errors: if the total normalized arithmetic perturbation is bounded by τ, the sufficient condition becomes $D/\Delta + \tau < 1/2$.

Common-pipeline check. The separate prototype replaces repeated filtering with (67)–(68). To prevent serial DNA maps from cancelling, each of the four rule streams selects a permutation on a different two-bit lane of each byte; decryption uses its inverse. The eight permutations are the first eight lexicographically ordered derangements of $(0, 1, 2, 3)$, representing four DNA symbols. Exhaustive checks verified all $8^4 = 4096$ byte maps and their inverses. The original XOR mask is retained only to isolate the functional effect of rule agreement.

PPO and the fixed-gain, LQR, and SMC baselines use the same nominal environment, initial seeds, input limits, and image pipeline, without retraining or retuning. The check uses a 128×128 version of the source image, 8×8 blocks, $\Delta = 1$, and 256 consecutive samples after 1,500 control intervals. Three independently trained PPO policies are each tested from three common initial conditions (nine trials per node); each deterministic baseline uses the same three initial conditions (three trials). No state guard was activated. Direct rounding and reconciliation use the same sender rules, substitutions, masks, and trajectories; only receiver rule reconstruction changes.

Table 15
Functional connection check, separate from the legacy image statistics. $D_{\max}$ is the largest four-stream error across trials. Mismatch is the mean rule mismatch percentage; recovery counts denote exact pixelwise recovery. D/R denotes direct rounding/reconciliation.

Node	Method	$D_{\max}$	Mismatch (%) D/R	Recovery D/R
2	PPO	0.1265	1.563/0	0/9; 9/9
2	Fixed	8.5900	73.307/70.671	0/3; 0/3
2	LQR	0	0/0	3/3; 3/3
2	SMC	0	0/0	3/3; 3/3
3	PPO	0.0360	0.347/0	0/9; 9/9
3	Fixed	0	0/0	3/3; 3/3
3	LQR	0	0/0	3/3; 3/3
3	SMC	0	0/0	3/3; 3/3

Table 15 shows that the tested PPO residuals satisfy the four-stream condition, yet direct rounding still creates mismatched rules. Reconciliation removes these mismatches and gives zero image error in all 18 PPO trials. The same mechanism also works for the successful traditional controllers; the displayed zero stream errors mean identical stored floating-point samples, not a mathematical finite-time synchronization theorem. Node 2 fixed-gain feedback fails the condition and does not recover the image in these tests. These results connect measured synchronization quality to discrete recovery without establishing a PPO-specific encryption advantage or a population reliability rate.

Scope and communication cost. This construction requires aligned auxiliary offsets; their compact JSON representation is approximately 20.7 kB per tested image, about 42% of the uncompressed image bytes. Their information leakage has not been bounded. The rule digest used to reject disagreements is unkeyed and is not authentication. Passing it is not proof of state synchronization: modulo-eight aliases can preserve the rules even outside (69). Correct recovery is therefore distinct from secrecy, tamper resistance, and secure key agreement. The conditional proposition does not validate the legacy preprocessing or establish cryptographic security for either image transform.

6. Conclusion

This study examines actuator placement and feedback-information availability in bounded single-input synchronization of a continuous-time Hopfield network. Its main finding is the actuator-dependent effect of observation composition: at a fixed four-dimensional input size, the alternative coordinate subset improves Node 2 but increases residuals and control effort at Node 3. This motivates considering actuator location together with available policy inputs within the studied setting, rather than treating low-residual synchronization at one node as transferable to another.

The complete actuator-subset comparison quantifies the cost of restricting actuation. Single-input Node 3 uses fewer actuators and less measured total effort than the favorable multi-input configurations, with a higher full-horizon error but not uniformly worse steady-state precision. Traditional-controller and SAC results show condition-dependent error–effort tradeoffs; they do not support a universal PPO advantage. Analytical state boundedness, finite-time conditional Lyapunov estimates, and multi-step sampled sensitivity checks provide complementary evidence, without constituting a formal synchronization or robustness guarantee.

The resulting contribution is a comparative assessment of actuator-dependent observation sensitivity and actuation-resource tradeoffs, supported by an explicit separation of analytical properties and numerical evidence. The accuracy-constrained selection study further links this assessment to an explicit resource preference and held-out initial-state checks, including no-selection outcomes. Its scope is one network and the tested policies and input subsets; the illustrative tolerances do not define application requirements or certify reliability. The tested choices are insensitive to the examined pass-fraction levels but can change with resource priority, and their original accuracy requirements cease to hold at the larger tested mismatch amplitudes. Future work should extend the observation-subset comparison, test whether the placement dependence persists under changed network parameters, and develop certified synchronization-error bounds for sampled learned feedback.

A separate image-reconstruction experiment makes the application requirement explicit: a sufficient four-stream error condition, together with aligned auxiliary offsets, guarantees discrete rule agreement and hence exact recovery under an identical mask. The common-pipeline tests verify this connection for the evaluated trajectories, rather than assuming that a small average synchronization error suffices.

This functional result does not establish cryptographic security. A final communication protocol requires additional treatment of auxiliary-information leakage, authentication, and key management.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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